

Curiosity-Driven Questioning for Engine-Agnostic LLM Research Ideation

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Abstract

Scientific ideation is driven by curiosity: researchers ask questions that expose knowledge gaps, reveal competing hypotheses, and clarify missing evidence relevant to decision making. Yet, most LLM-based ideation systems optimize the *idea text* while leaving curiosity under-modeled, resulting in brittle, engine-specific gains. To address this, we propose **Curiosity-Driven Questioning (CDQ)**, a training-free, engine-agnostic method that constructs a compact set of curiosity-driven questions from a topic corpus under black-box LLM access. CDQ scores questions with three corpus-relative indices (GAPEXPOSURE, DISAGRDIVERSITY, CLARITYGAIN) and uses a two-stage scoring with an entropic mirror-ascent reweighting over question meta-policies to concentrate on high-utility question policies. Our results show that prepending CDQ questions improves downstream ideation across single-pass, retrieval-augmented generation, and self-reflection engines with multiple base models under preference-based evaluation. Ablations and a human study further support the complementarity of the indices and the reliability of the evaluation. Our code and dataset are available at: <https://github.com/Sikun-Skyler-Guo/CDQ>

CCS Concepts

• Computing methodologies → Natural language processing.

Keywords

AI4Science, Large Language Models, Inference-time Learning

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1 Introduction

Large language models (LLMs) are increasingly used as scientific assistants: they can synthesize large volumes of papers, draft technical text, and even participate in end-to-end discovery workflows [7, 8, 17, 18, 29, 31]. Yet, *research ideation*, turning a topic-specific snapshot of the literature into a small set of actionable research directions, remains brittle: without careful scaffolding, LLM-generated ideas are often plausible, but shallow or weakly grounded [16, 36]. A key reason is that ideation is not only analysis, but also *active inquiry*: progress depends on deciding what to ask next of the literature and what missing evidence would change the potential findings. To behave like human researchers, LLM assistants must therefore do more than summarize and draft text: they must exhibit *curiosity* by proactively formulating high-quality questions to interrogate a literature snapshot. Without an explicit questioning step, even strong generators tend to default to generic “future work” patterns, producing ideas that sound plausible but do not target the decision-relevant information.

Human researchers rarely ideate by one-shot enumeration. Instead, they leverage curiosity to expose gaps, reveal competing hypotheses, and seek information relevant to decisions [5, 13, 20, 25, 26, 28]. This suggests *that good ideas are often downstream of good questions asked relative to the current literature*. For example, under a hallucination-mitigation topic, a generic question like “How can we reduce hallucinations in LLMs?” provides little guidance. In contrast, curiosity-driven questions can directly target corpus-relative unknowns, evidence-dependent disagreements, and the specific missing measurement or controlled comparison that would change the NOVELTY / FEASIBILITY of a proposed mitigation.

However, *quantifying* whether a question is “good” is nontrivial: many “unknowns” are vague, untestable, or irrelevant. Moreover, under black-box access (no embeddings or token log-probabilities) to LLMs, likelihood-based confidence signals are unavailable, pushing us toward lightweight and interpretable text-based criteria.

Most LLM ideation engines optimize the *idea text* directly, via retrieval-augmented grounding [3, 22, 23, 27, 39, 41] or iterative critique/refinement [10, 24, 30, 35, 37]. These advances improve

how an idea is generated or refined, but they largely leave *what to ask next of the literature* implicit, despite question quality being the main driver of effective scientific exploration. The inquiry step in previous work is often left implicit (e.g., fixed “open problems” prompts), making improvements brittle and difficult to transfer across engines. This motivates a plug-and-play “curiosity module”: a reusable way to construct a compact question set that can sit *in front of* any ideation engine and reliably steer exploration. Therefore, we raise our research question: **can we, at inference time and without model internals, optimize how we ask questions so that the resulting questions reliably improve downstream ideation across engines?**

We propose **Curiosity-Driven Questioning (CDQ)**, a *training-free* and *engine-agnostic* framework that constructs a compact set of research questions from a topic corpus $\mathcal{P}$. CDQ scores candidate questions with three complementary corpus-relative indices: (i) **GAPEXPOSURE** (does the question expose a concrete knowledge gap?), (ii) **DISAGRDIVERSITY** (does the question admit multiple incompatible yet plausible answers under the same evidence?), and (iii) **CLARITYGAIN** (does the question reduce uncertainty about decision-relevant levers for the target $y \in \{\text{NOVELTY, FEASIBILITY}\}$). Using a two-stage scoring pipeline, CDQ *iteratively reweights* a small library of human-inspired question meta-policies via an entropic mirror-ascent update, concentrating generation on question policies that yield higher curiosity utility. All scoring is computed under a strict corpus-only answering constraint, so curiosity is measured relative to the literature snapshot rather than parametric model knowledge.

We evaluate CDQ on 500 topics constructed from IdeaBench [16]. Prepending CDQ questions improves downstream ideation across three common engines (single-pass, retrieval-augmented generation (RAG), self-reflection) and multiple base models under a pairwise LLM-as-a-judge preference protocol, with ablations and a human study further supporting the complementarity of our indices and the reliability of the evaluation.

Our main contributions are:

- We formulate inference-time, black-box, engine-agnostic optimization of a corpus-grounded *questioning module* for research ideation.
- We introduce three interpretable curiosity indices, **GAPEXPOSURE**, **DISAGRDIVERSITY**, and **CLARITYGAIN**, that operationalize curiosity *relative to the current literature* and to the target dimension (**NOVELTY** or **FEASIBILITY**).
- We propose a training-free inference-time optimization paradigm: entropic mirror ascent over a mixture of question meta-policies that converges in a small number of rounds under strict iteration budgets.

2 Related Work

2.1 Curiosity and Question Asking in Human Research

Curiosity is widely viewed as a driver of exploration and knowledge acquisition. Foundational accounts emphasize novelty/complexity/incongruity as triggers for exploratory cognition [5, 6], while the information-gap theory models curiosity as the tension induced by

recognizing a specific missing piece of understanding [28]. More recent cognitive neuroscience and computational perspectives frame curiosity as an information-seeking policy that balances intrinsic uncertainty reduction with instrumental, decision-relevant value [13, 20]. Psychometric work also distinguishes diverse (broad) from specific (gap-closing) epistemic curiosity [25, 26]. CDQ is directly inspired by these views and operationalizes curiosity *relative to a literature snapshot* via gap exposure, hypothesis branching, and decision-relevant uncertainty reduction for novelty/feasibility.

2.2 LLMs for Research Ideation

Retrieval-augmented and literature-grounded ideation. A central requirement for ideation is grounding: ideas should be novel *relative to* the literature while remaining evidence-consistent. RAG-style generation [22] and literature-based discovery [38] motivate systems that retrieve papers to steer hypothesis and direction generation. Recent frameworks use retrieval and structured access to scientific corpora to propose and refine research directions [3, 23, 27, 39, 41], with benchmarks such as IdeaBench enabling large-scale evaluation [16]. However, retrieval does not by itself specify *what questions to ask* of the retrieved evidence; in practice, many systems rely on fixed “open problems/future work” prompts, which can yield generic or miscalibrated exploration.

Self-reflection and iterative reasoning for ideation. Another line improves ideation by iterative critique and refinement through self-feedback, tool-use, or multi-agent deliberation [15, 30, 35, 42, 43]. Debate and discussion have been used to encourage divergent thinking and reduce obvious errors [10, 24], and recent multi-agent ideation systems further mimic collaborative scientific workflows [37]. These methods can expand the search space, but they may still drift when critique is not grounded in the same literature snapshot that defines novelty and feasibility.

Our position. CDQ is complementary to both families: instead of prescribing a particular ideation engine, we optimize a compact *front-end question set* that any downstream engine can condition on. This separation makes the approach engine-agnostic: a RAG pipeline can use CDQ questions as discriminative information requests, while a self-reflection pipeline can use them to structure critique around corpus-relative gaps and decision-relevant levers.

3 Method

3.1 Problem Setup

Inference-time curiosity optimization. Given a topic t with a corpus $\mathcal{P}_t$ (titles/abstracts), our goal is to output a compact question set $\mathcal{Q}_t = \{q_k\}_{k=1}^K$, consisting of K high-value questions (q_k denotes the k -th individual question), to improve downstream research ideation along a target $y \in \{\text{NOVELTY, FEASIBILITY}\}$. We assume *black-box* LLM access: no token probabilities or embeddings are available. CDQ is *engine-agnostic*: it optimizes questions without committing to a specific ideation engine (single-pass, RAG, self-reflection, etc.). Candidate questions are scored by three corpus-relative curiosity indices: **GAPEXPOSURE** (knowledge gaps), **DISAGRDIVERSITY** (branching hypotheses), and **CLARITYGAIN** (decision relevance for y).

LLM agent roles. We instantiate three agents from the same underlying LLM whose templates are listed in the Appendix A.1.2. In

addition, to remain within context budgets, all calls are conditioned on a compact evidence packet retrieved from $\mathcal{P}_t$, and all outputs are required to be grounded in that packet (see Appendix A.1.2 for details). The three agents are:

- **Generator G** : proposes candidate questions given a meta-policy instruction and topic context evidence packet.
- **Evaluator E** : produces short *evidence-grounded* answers (and, for clarity, short idea drafts) using only the evidence packet.
- **Judge J** : outputs one-token labels (answerability, answer relations, lever labels) used to compute the indices.

3.2 Three Complementary Curiosity Indices

Each candidate question q is scored by GAPEXPOSURE, DISAGRDIVERSITY, and CLARITYGAIN. We use S_g, S_d, S_c to denote the numbers of samples used to estimate these indices, respectively.

3.2.1 GAPEXPOSURE: does q expose a knowledge gap in $\mathcal{P}$? A good question should help expose a knowledge gap in $\mathcal{P}$. Given q , the evaluator returns a short answer *a from the evidence packet* and an answerability tag $\tau \in \{\text{Answerable, Partial, Unknown}\}$.

Tags’ operational meaning.

- **Answerable**: sufficient evidence supports a complete and specific answer.
- **Partial**: some evidence exists, but key details are missing.
- **Unknown**: insufficient evidence; answering q would require information outside $\mathcal{P}$.

Tag-to-score mapping. Based on tags’ operational meaning, we map tags to per-sample gap scores: $g(\text{Answerable})=0, g(\text{Partial})=\frac{1}{2}$, and $g(\text{Unknown})=1$.

We draw S_g i.i.d. samples $\{(a_s, \tau_s)\}_{s=1}^{S_g} \sim E(\cdot \mid q, \mathcal{P})$ and compute the empirical distribution

$$\hat{\rho}_q(\tau) := \frac{1}{S_g} \sum_{s=1}^{S_g} \mathbf{1}[\tau_s = \tau] \quad (1)$$

over the S_g answerability tags, where $\mathbf{1}[\cdot]$ denotes the indicator function. We score GAPEXPOSURE by the empirical expected gap score:

$$\begin{aligned} \text{Gap}(q) &:= \sum_{\tau} \hat{\rho}_q(\tau) g(\tau) \\ &= \hat{\rho}_q(\text{Unknown}) + \frac{1}{2} \hat{\rho}_q(\text{Partial}) \end{aligned} \quad (2)$$

High $\text{Gap}(q)$ indicates that the topic literature is insufficient to answer q .

Efficiency (optional proxy gate). We optionally use a cheap judge-based proxy to skip full estimation for clearly high-gap questions [4]; and details are in the Appendix A.1.2.

3.2.2 DISAGRDIVERSITY: does q reveal competing hypotheses under $\mathcal{P}$? A useful question is not merely unanswered; it should induce *branching* plausible answers under the same evidence. Given question q and the corpus $\mathcal{P}$, we sample S_d corpus-grounded short answers $\{a_1, \dots, a_{S_d}\} \sim E(\cdot \mid q, \mathcal{P})$. For each answer pair (a_i, a_j) , the judge outputs a one-token relation label:

$r_{ij} \in \{\text{Entails, Contradicts, Unrelated}\}$.

Relation labels’ operational meaning.

- **Entails**: same semantic answer type (We treat it as a symmetric merge relation).
- **Contradicts**: cannot both be true simultaneously under a consistent interpretation (competing hypotheses).
- **Unrelated**: neither equivalence nor contradiction is supported, given the evidence.

Clustering and disagreement entropy. We merge logically consistent answers via entailment connectivity (using a union–find algorithm over Entails edges), which partitions the S_d samples into C disjoint clusters $\{C_c\}_{c=1}^C$. Let $\hat{f}_c := |C_c|/S_d$ be the empirical frequency of the c -th cluster. We then define DISAGRDIVERSITY as the Miller–Madow corrected entropy:

$$\text{Disagr}(q) := - \sum_{c=1}^C \hat{f}_c \log \hat{f}_c + \frac{C-1}{2S_d}. \quad (3)$$

High $\text{Disagr}(q)$ indicates multiple distinct plausible answer types under the same evidence, signaling questions that can drive discovery by adjudicating between mechanisms or regimes.

3.2.3 CLARITYGAIN: does q clarify decision-relevant levers for target y ? Curiosity is most valuable when it is *decision-relevant*: potential answers should reduce the uncertainty about why a potential idea is novel or feasible. We operationalize this using an interpretable lever panel inspired by prior ideation benchmarks and systems [2, 16, 40, 44].

Lever panel Z_y . For each target $y \in \{\text{NOVELTY, FEASIBILITY}\}$, we define the levers $M=8$ $Z_y = (Z_{y,1}, \dots, Z_{y,M})$. Intuitively, each lever corresponds to a specific decision-relevant aspect along which a question or idea may reduce the uncertainty of the target y , such as clarifying assumptions, mechanisms, constraints or evaluation criteria. The judge labels each lever as $v \in \mathcal{V} := \{\text{Yes, No, Uncertain}\}$, indicating whether the current context provides sufficient evidence for that aspect. More detailed lever definitions and prompts are provided in the Appendix A.1.2.

Entropy given a context. Given a context x (evidence packet only, or evidence packet plus question/idea draft), for each lever $Z_{y,m}$ we repeat the judge labeling N times, obtaining labels $\{v_m^{(n)}\}_{n=1}^N$ with $v_m^{(n)} \in \mathcal{V}$. This induces an empirical distribution over $v \in \mathcal{V}$:

$$\hat{p}_m(v \mid x) := \frac{1}{N} \sum_{n=1}^N \mathbf{1}[v_m^{(n)} = v], \quad v \in \mathcal{V}. \quad (4)$$

We define the average lever entropy:

$$H_y(x) := \frac{1}{M} \sum_{m=1}^M \left(- \sum_{v \in \mathcal{V}} \hat{p}_m(v \mid x) \log \hat{p}_m(v \mid x) \right). \quad (5)$$

Baseline entropy. With *corpus-only* context $x = \mathcal{P}_t$ (evidence packet without asking q), we compute baseline entropy $H'_y := H_y(\mathcal{P}_t)$ by repeating judge labeling N_{baseline} times.

Post-question entropy via idea drafts. For a question q , we sample S_c idea drafts $\{i_s\}_{s=1}^{S_c}$ from the evaluator (each includes idea and grounding). With context $x = (\mathcal{P}_t, q, i_s)$, we compute the post-question entropy $H_y^{(q,i_s)} := H_y(\mathcal{P}_t, q, i_s)$ by repeating the judge labeling $N_{\text{post-question}}$ times.

Clarity gain. We define CLARITYGAIN as the expected entropy reduction:

$$\begin{aligned} \text{Clarity}_y(q) &:= \mathbb{E}_i \left[H'_y - H_y^{(q,i)} \right] \\ &\approx \frac{1}{S_c} \sum_{s=1}^{S_c} \left(H'_y - H_y^{(q, i_s)} \right). \end{aligned} \quad (6)$$

Intuitively, $\text{Clarity}_y(q)$ is high when asking q yields grounded drafts that pin down which levers matter for y ; and it admits an information-gain interpretation. CLARITYGAIN measures how much asking a question reduces uncertainty over a decision-relevant lever panel for NOVELTY/FEASIBILITY. Concretely, if the corpus alone leaves multiple levers ambiguous (e.g., unclear evaluation protocol, unknown data availability, unspecified confounds), a high CLARITYGAIN question induces grounded drafts where these levers become consistently resolvable (Yes/No rather than Uncertain), thereby lowering lever entropy. This makes the ideation engine more likely to produce actionable and testable directions rather than generic future work. More theoretical analysis on the decision-theoretic meaning of CLARITYGAIN can be found in the Appendix A.2.

Why do all three indices matter. GAPEXPOSURE filters the already-answered questions; DISAGRDIVERSITY prefers branching questions; CLARITYGAIN enforces the relevance of the decision by measuring the reduction in lever-uncertainty.

3.3 Utility, two-stage scoring, and inference-time learning over meta-policies

Previous work in cognitive science and causal reasoning suggests that curiosity-driven questions arise from qualitatively different inquiry operators, such as identifying information gaps, drawing analogies across domains, probing counterfactuals and boundary conditions, and reflecting on known failures [11, 21, 28, 30, 33, 35]. Similarly, recent LLM-based ideation systems indicate that no single heuristic consistently produces high-quality research questions across domains or objectives [2, 40].

We formulate a small library of human-inspired question meta-policies $\Pi = \{\pi_1, \dots, \pi_5\}$ that captures complementary modes of scientific inquiry that are difficult to elicit reliably from a single prompting strategy. Concretely, Π includes:

- π_1 (**Info-gap-hunter**): identify missing evidence and what observation would close the gap.
- π_2 (**Analogy-bridger**): propose cross-domain transfer and ask when the analogy breaks.
- π_3 (**Counterfactual-boundary**): stress-test assumptions via regime/counterfactual shifts.
- π_4 (**Failure-to-question**): turn known limitations into discriminative follow-up questions.
- π_5 (**Constraint-eval-reframer**): focus on constraints, evaluation, and measurable comparisons.

Prompt templates for the meta-policies are in the Appendix A.1.2. **Mixture sampling.** We run CDQ for a small number of rounds $\ell = 1, 2, \dots, \mathcal{L}$. In round ℓ , we sample L candidate questions from a mixture on the library of meta-policies Π with weights $\mathbf{w}^{(\ell)} \in \Delta^{|\Pi|}$, where

$$\Delta^{|\Pi|} := \left\{ \mathbf{w}^{(\ell)} \in \mathbb{R}^{|\Pi|} \geq 0 : \sum_{i=1}^{|\Pi|} w_i^{(\ell)} = 1 \right\}. \quad (7)$$

Since $|\Pi| = 5$, $\mathbf{w}^{(\ell)}$ is a 5-dimensional probability vector on meta-policies. To ensure exploration, we first generate one candidate question from each π_i , then sample the remaining $L - |\Pi|$ candidates by drawing $\pi \sim \text{Cat}(\mathbf{w}^{(\ell)})$ and prompting the generator with that policy. Here, $\text{Cat}(\mathbf{w}^{(\ell)})$ denotes a categorical distribution with probabilities $\mathbf{w}^{(\ell)}$.

Two-stage scoring. Computing $\text{Gap}(q)$ and $\text{Disagr}(q)$ is cheap, while $\text{Clarity}_y(q)$ is expensive since it involves idea drafts and repeated lever labeling. We therefore compute $\text{Gap}(q)$ and $\text{Disagr}(q)$ for all candidates, keep the top r_{surv} survivors by $\alpha \text{Gap}(q) + \beta \text{Disagr}(q)$, and compute $\text{Clarity}_y(q)$ only for survivor questions.

Per-question utility. Within each round, we min-max normalize each index to $[0, 1]$ across the round's candidates/survivors, and combine them:

$$U_y(q) := \alpha \text{Gap}(q) + \beta \text{Disagr}(q) + \gamma \text{Clarity}_y(q), \quad (8)$$

where (α, β, γ) are hyperparameters that control the relative importance of the indices.

Mirror-ascent update. Let $\mathcal{S}^{(\ell)}$ be the survivor set at round ℓ , and $\mathcal{S}_i^{(\ell)} \subseteq \mathcal{S}^{(\ell)}$ the survivors generated by π_i . We summarize policy quality by the median survivor utility:

$$\bar{U}_y^{(\ell)}(i) := \text{median}\{U_y(q) : q \in \mathcal{S}_i^{(\ell)}\}, \quad (9)$$

and set $\bar{U}_y^{(\ell)}(i) = 0$ if $\mathcal{S}_i^{(\ell)} = \emptyset$. We update the mixture via an exponentiated-gradient (entropic mirror-ascent) step:

$$w_i^{(\ell+1)} = \frac{w_i^{(\ell)} \exp\{\eta \bar{U}_y^{(\ell)}(i)\}}{\sum_{j=1}^{|\Pi|} w_j^{(\ell)} \exp\{\eta \bar{U}_y^{(\ell)}(j)\}}. \quad (10)$$

Here $\eta > 0$ is the learning rate, controlling how aggressively the mixture concentrates on high-utility meta-policies. This update is the closed-form solution to a KL-regularized proximal objective on the simplex: it increases the expected utility under $\bar{U}_y^{(\ell)}$ while penalizing deviation from the previous mixture. Consequently, meta-policies that yield higher-utility survivors receive more weight over rounds. It is the standard multiplicative-weights / entropic mirror-ascent update with no-regret guarantees [1]. Additional theoretical analyses of CDQ are provided in the Appendix A.2.

Stopping and output. We run at most five rounds and early-stop when the round-level median survivor utility fails to improve. The final output consists of the top- K questions selected from all generated candidates, ranked by their specific utility scores $U_y(q)$. In cases where utility scores are tied, we prioritize questions with higher Clarity_y .

Figure 1 depicts the CDQ execution flow, which operates as an iterative inference-time optimization loop. First, evidence packets are constructed from the corpus with respect to the target y . Candidate questions are then generated through meta-policies and evaluated through a two-stage scoring process: Stage 1 filters survivors using $\text{Gap}(q)$ and $\text{Disagr}(q)$, while Stage 2 calculates the final utility $U_y(q)$ via $\text{Clarity}_y(q)$. To maximize inquiry quality, the

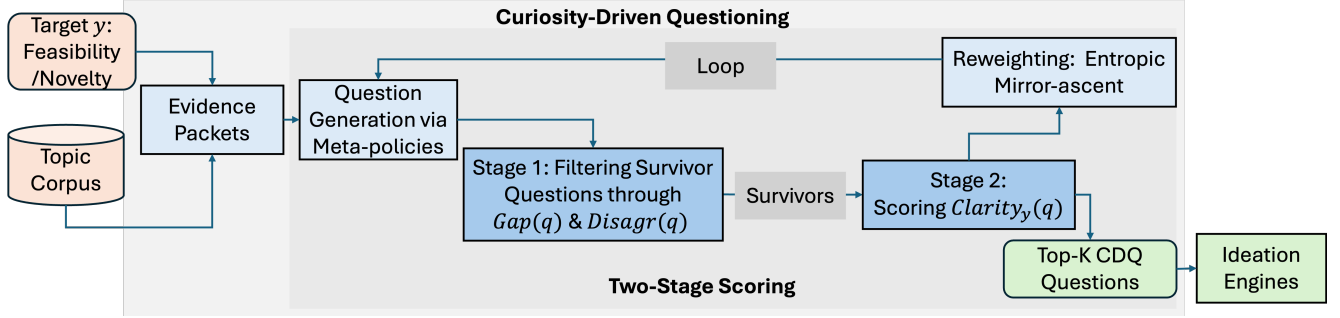


Figure 1: CDQ pipeline: We construct evidence packets using topic corpus with respect to target y . Then we generate questions via meta-polices based on the evidence packets. Next, the questions go through our two-stage scoring process, where stage 1 filters survivor questions using $Gap(q)$ and $Disagr(q)$; and stage 2 scores the survivor questions using $Clarity_y(q)$ to get the final question utility $U_y(q)$. To maximize question utility $U_y(q)$, we update the weights for meta-polices via entropic mirror-ascent, and output top- K CDQ questions for downstream ideation engines.

meta-policy weights are updated via entropic mirror-ascent, with the top- K questions finally output to downstream ideation engines.

More details are implemented in Algorithm 1 in Appendix A.1.2.

4 Experiments

4.1 Experiment Setup

Data. We follow the topic-corpus format of Guo et al. [16] and construct a dataset with 500 topics. Each topic t is defined by a target paper and a topic-specific corpus $\mathcal{P}_t$ consisting of titles/abstracts from its reference papers. To fit token budgets, all CDQ components and ideation engines operate on a compact *evidence packet* derived from $\mathcal{P}_t$.

Question learning with CDQ. For each topic t and target dimension $y \in \{\text{NOVELTY}, \text{FEASIBILITY}\}$, we run CDQ (Algorithm 1 in the Appendix A.1.2) to produce a set of questions $\mathcal{Q}_t^{(y)}$. Unless otherwise stated, CDQ runs for at most 5 rounds with 30 candidate questions per round, keeps 15 survivors for the expensive CLARITYGAIN stage, and outputs the top- $K = 10$ questions for the downstream ideation engine. We use $S_g = 3$ samples for GAPEXPOSURE, $S_d = 5$ samples for DISAGRDIVERSITY, and $S_c = 5$ idea drafts for CLARITYGAIN, with utility weights $(\alpha, \beta, \gamma) = (0.2, 0.3, 0.5)$ and EG step size $\eta = 1.0$. We run CDQ separately for NOVELTY and FEASIBILITY because CLARITYGAIN depends on y .

Downstream ideation engines. To test whether CDQ transfers across ideation pipelines, we evaluate CDQ with three representative engines, each under two conditions: *Baseline* (corpus only) and *CDQ augmentation* (corpus + top- $K=10$ CDQ questions ranked by U_y).

- **Single-pass:** generate one research idea in a single prompt (with or without $\mathcal{Q}_t^{(y)}$).
- **RAG:** retrieve additional context (e.g., via web search) and then synthesize an idea; in the CDQ condition, CDQ questions are used to steer retrieval.
- **Iterative self-reflection:** iteratively critique and revise an initial idea for a fixed number of steps; in the CDQ condition, questions are used as a structured reflection checklist.

Prompt templates and implementation details for all engines are provided in the Appendix A.1.2.

4.2 Evaluation: LLM-judge win-rate (ideas with vs. without CDQ)

We evaluate whether CDQ’s questions lead to *better final ideas* than running the same ideation engine without CDQ.

What we measure (win-rate). Our primary metric is the win-rate: the fraction of topics for which a judge prefers the CDQ-augmented idea over the baseline idea for a given target $y \in \{\text{NOVELTY}, \text{FEASIBILITY}\}$. A win-rate greater than 50% indicates that CDQ improves ideation more often than not.

Pairwise protocol. For each topic t and engine e , we generate two final ideas: $I_{t,e}^+$ (with CDQ questions), and $I_{t,e}^0$ (baseline).

We then present to the LLM judge with (i) the same evidence packet, (ii) the target y , and (iii) the two anonymized ideas in randomized order. The LLM judge is presented with the two ideas labeled simply as Option A and Option B. The judge must make a forced choice $\{A, B\}$ based on which idea better satisfies the target y . Because LLM judges can be stochastic, we repeat each comparison $M_{\text{eval}}=5$ times with independently re-randomized A/B order. Let $J_{t,e,m} \in \{A, B\}$ be the decision on repetition m . We define the final winner by majority vote:

$$\widehat{W}_{t,e} := \arg \max_{c \in \{A, B\}} \sum_{m=1}^{M_{\text{eval}}} \mathbf{1}[J_{t,e,m} = c]. \quad (11)$$

Since M_{eval} is odd, ties cannot occur.

Win-rate computation. Let N_{eval} be the number of topics evaluated, and let $\sigma_{t,e}(\text{CDQ}) \in \{A, B\}$ indicate which side corresponds to $I_{t,e}^+$ after randomization. We report

$$\text{win-rate} := \frac{1}{N_{\text{eval}}} \sum_{t=1}^{N_{\text{eval}}} \mathbf{1}[\widehat{W}_{t,e} = \sigma_{t,e}(\text{CDQ})], \quad (12)$$

calculated separately for NOVELTY and FEASIBILITY and separately for each engine e . We follow standard mitigation practices for LLM-based evaluation: repeated judging with majority vote and randomized presentation order [14, 19]. Across baseline and CDQ augmentation, we (i) use the same ideation budget, (ii) keep retrieval

Table 1: Win-rate (%) of CDQ-augmented ideas vs. baseline across base models and ideation engines (forced-choice judge, $M_{\text{eval}}=5$ repeats + majority vote). Bold and underlined values denote the highest and lowest scores for each metric, respectively.

Base Model	Ideation Engine	Novelty	Feasibility
GPT-4o	Single-pass	70.80	77.80
	RAG	73.00	89.20
	Self-reflection	54.80	74.40
Qwen	Single-pass	95.40	<u>71.00</u>
	RAG	76.80	91.00
	Self-reflection	66.80	68.80
Gemini	Single-pass	91.38	88.00
	RAG	73.74	96.80
	Self-reflection	74.80	86.00
Minimax-m2	Single-pass	72.40	76.00
	RAG	67.80	91.80
	Self-reflection	<u>51.40</u>	89.80

settings fixed, and (iii) blind the judge to condition labels via A/B randomization on every repetition.

Results and findings. Table 1 reports win-rates in four base models (GPT-4o¹, Qwen, Gemini and Minimax-m2) and three ideation engines. Table 2 further summarizes robustness across research discipline clusters by macro-averaging win-rates within each cluster (averaged across base models). The findings are as follows:

- **CDQ improves ideation across models and engines.** CDQ achieves win-rates $> 50\%$ for *all* model-engine combinations on both NOVELTY and FEASIBILITY (Table 1), supporting engine-agnostic transfer.
- **CDQ pairs especially well with RAG for FEASIBILITY.** The RAG feasibility win-rates are consistently high in all base models (Table 1), suggesting that CDQ questions act as discriminative information requests that steer the retrieval towards implementation and evidence relevant to evaluation.
- **Single-pass gains are strongest for NOVELTY while self-reflection favors FEASIBILITY.** Single-pass sees particularly large novelty improvements for some base models, while iterative self-reflection tends to yield stronger feasibility win-rates but smaller novelty gains (Table 1).
- **Robustness across scientific disciplines.** CDQ yields a win-rate greater than 50% in all clusters of disciplines for both targets (Table 2).

Vote-margin stability. To further check whether majority-vote outcomes are driven by brittle LLM-judge randomness, we inspect the vote margins across the five independently randomized judge repetitions for each comparison. In 72% of comparisons, four out of the five repeated judge calls selected the same option, while only one selected the other option. This suggests that the majority-vote

result is usually based on a clear preference pattern rather than a narrow or unstable 3–2 split.

4.3 Ablation Study

We ablate CDQ components to disentangle (i) the contributions of the three curiosity indices and (ii) the benefit of inference-time learning over the meta-policy mixture. To keep the study lightweight, we run GPT-4o with *single-pass* ideation on a random subset of 100 topics. All variants are evaluated using the identical repeated forced-choice protocol in Sect. 4.2 ($M_{\text{eval}} = 5$ with majority vote), and we keep the downstream ideation prompt format and $K = 10$ prepended questions fixed across variants.

Variants. We compare Full CDQ to the following controlled ablations:

- w/o GAPEXPOSURE, w/o DISAGRDIVERSITY, and w/o CLARITYGAIN, where we set the corresponding term’s utility weight to zero while keeping the rest of the pipeline unchanged (two-stage scoring and policy reweighting remain active).
- (ii) No mirror update, which runs only the first CDQ round with a *uniform* meta-policy mixture $w = 1/|II|$ and outputs the top- K questions from that round. This variant keeps the two-stage scoring pipeline, and therefore tests whether a fixed, non-adaptive mixture of meta-policies is sufficient.
- (iii) Pure-random question, which removes both mirror-ascent reweighting and two-stage scoring. Under the same question budget, it samples topic-conditioned candidate questions from the uniform meta-policy mixture and randomly selects the final top- K questions without using GAPEXPOSURE, DISAGRDIVERSITY, or CLARITYGAINSCORES.
- (iv) Generic-Q, a domain-agnostic checklist baseline that prepends a fixed set of generic questions that do not depend on the topic corpus (examples include: “What is the exact hypothesis/claim?”; “What dataset or evidence would be required, and is it available?”; “What is the evaluation metric/protocol?”; “What baselines should be compared against?”; “What assumptions are critical, and what confounders could break them?”; “What would falsify the hypothesis?”).

Results and insights. Table 3 shows three consistent patterns.

(1) **All indices are complementary and target-sensitive.** Removing any index degrades win-rates, with effects that align with each index’s intended role. GAPEXPOSURE is most critical for NOVELTY (73%→59%) by steering questions toward corpus-relative unknowns rather than already-answered themes, while DISAGRDIVERSITY is particularly important for FEASIBILITY (79%→58%), reflecting the value of surfacing mutually incompatible yet plausible implementation paths that must be adjudicated. CLARITYGAIN is necessary for both targets and produces the largest overall drop for NOVELTY (73%→55%), supporting that decision-relevant uncertainty reduction is central to producing actionable, non-generic directions.

(2) **Adaptive concentration, rather than random question addition or one-shot scoring alone, explains the remaining gain.** Pure-random question control obtains 63% on NOVELTY and 64% on FEASIBILITY, which is nearly indistinguishable from No mirror update (62% / 64%). Thus, although No mirror update still

¹ Model versions: gpt-4o-2024-08-06 via OpenAI; google/gemini-2.5-flash, Qwen/Qwen3-Next-80B-A3B-Instruct, and MiniMaxAI/MiniMax-M2 are accessed via DeepInfra endpoints.

Table 2: Win-rate (%) of CDQ-augmented ideas vs. baseline across discipline clusters, averaged across base models (GPT-4o, Qwen, Gemini, Minimax-m2). OVERALL is the unweighted (macro) average across clusters. Bold and underlined values denote the highest and lowest scores for each metric, respectively.

Discipline Cluster	Novelty win-rate			Feasibility win-rate		
	Single-pass	RAG	Self-reflection	Single-pass	RAG	Self-reflection
Environmental Sciences	82.48	73.00	63.50	81.00	89.63	65.51
Genetics and Mol. Biology	78.00	73.25	<u>58.50</u>	78.50	91.87	68.12
Health and Medicine	81.25	73.75	63.20	79.25	87.12	67.18
Neuroscience and Cog. Sci.	86.50	71.50	64.25	76.25	89.64	64.49
Technology and Engineering	84.25	72.00	60.25	75.50	87.73	<u>63.67</u>
OVERALL	82.50	72.70	61.94	78.10	89.18	65.79

Table 3: Ablation on a 100-topic subset (single-pass gpt-4o). Win-rates use the repeated forced-choice protocol in Sect. 4.2 ($M_{\text{eval}}=5$ + majority vote).

Variant	NOVELTY	FEASIBILITY
Full CDQ	73%	79%
w/o GAPExPOSURE	59%	70%
w/o DISAGRDIVERSITY	67%	58%
w/o CLARITYGAIN	55%	61%
No mirror update	62%	64%
Pure-random question	63%	64%
Generic-Q	48%	51%

uses the two-stage scoring pipeline, its performance is not meaningfully better than randomly selecting topic-conditioned questions from a uniform meta-policy mixture. In contrast, Full CDQ outperforms both controls by roughly 10 to 11 points on NOVELTY and 15 points on FEASIBILITY, suggesting that the benefit comes from adaptively concentrating question generation toward higher-utility meta-policies over rounds.

(3) **Gains are not from generic “checklists” alone.** Generic-Q performs near chance (48% / 51%), suggesting that simply prepending generic reflective questions provides little benefit. In contrast, both No mirror update and Full CDQ substantially outperform Generic-Q, indicating that corpus-relative scoring and adaptive policy concentration are necessary to obtain consistently useful questions for ideation.

4.4 Interpreting Mirror-Ascent Meta-Policy Learning

To provide interpretable evidence that CDQ’s inference-time learning is doing more than simply prepending questions, we inspect the *final* meta-policy mixture weights produced by the entropic mirror-ascent update in Eq. 10. For each topic, we extract the learned weight vector $w^{(L)} \in \Delta^5$ and macro-average within each of the five research discipline clusters ($n = 100$ per cluster). Figure 2 visualizes these averages as stacked bars for both NOVELTY and FEASIBILITY targets.

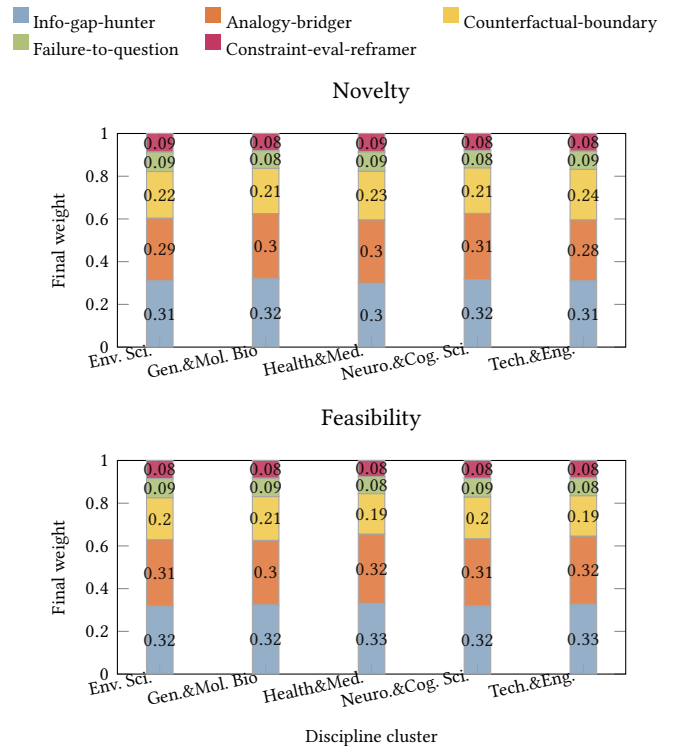


Figure 2: Final meta-policy mixture weights learned by entropic mirror-ascent (Eq. 9), averaged over topics in each discipline cluster ($n = 100$ per cluster).

Where the learning concentrates. Across all clusters, mirror-ascent consistently allocates most probability mass to INFO-GAP-HUNTER and ANALOGY-BRIDGER (each typically ≈ 0.29 – 0.33), while assigning moderate weight to COUNTERFACTUAL-BOUNDARY (≈ 0.19 – 0.24) and smaller weights to FAILURE-TO-QUESTION and CONSTRAINT-EVAL-REFRAMER (≈ 0.075 – 0.092). This pattern is intuitive: gap-seeking and analogy-based meta-policies generate questions that most often surface corpus-relative unknowns and transferable mechanisms, while counterfactual probing remains useful as a complementary stress-test operator.

Table 4: Entropy of the final meta-policy weight distribution w (nats; $H_{\max} = \log 5 = 1.609$).

Cluster	$H(w)$	$H(w)$
	Novelty	Feasibility
Environmental Sciences	1.486	1.473
Genetics & Molecular Biology	1.465	1.470
Health & Medicine	1.487	1.444
Neuroscience & Cognitive Sciences	1.463	1.469
Technology & Engineering	1.478	1.457
OVERALL (macro avg.)	1.476	1.462

Target-dependent shifts. Comparing the two targets, FEASIBILITY shows a mild but consistent shift toward INFO-GAP-HUNTER and ANALOGY-BRIDGER, with reduced emphasis on COUNTERFACTUAL-BOUNDARY. For instance, in Health & Medicine the FEASIBILITY mixture places ≈ 0.33 on INFO-GAP-HUNTER and ≈ 0.32 on ANALOGY-BRIDGER, but only ≈ 0.19 on COUNTERFACTUAL-BOUNDARY. This supports that feasibility-oriented questioning prioritizes actionable evidence requirements and concrete evaluability over broader hypothesis branching.

Quantifying concentration via weight entropy. We quantify how far the learned mixture deviates from uniform sampling using Shannon entropy $H(w) = -\sum_{i=1}^5 w_i \log w_i$ (nats), where uniform weights attain the maximum $H_{\max} = \log 5 = 1.609$. As shown in Table 4, the learned mixtures have $H(w)$ in the range 1.44–1.49 (macro-average: 1.476 for NOVELTY, 1.462 for FEASIBILITY). This indicates a clear departure from uniform sampling, yet the entropy remains far from zero, suggesting that mirror-ascent concentrates probability mass toward higher-utility meta-policies without collapsing to a single meta-policy. This behavior matches the entropic mirror-ascent update in Eq. 10, which increases expected utility while penalizing overly aggressive deviation, helping robustness under noisy utility estimates across disciplines.

Effective number of meta-policies. To make this concentration more interpretable, we report the *effective number of meta-policies* $\exp(H(w))$, which can be read as the number of policies that meaningfully contribute under the mixture. With $|\Pi| = 5$, a uniform mixture of weights corresponds to $\exp(H) = 5$. In contrast, our learned mixture of weights yield $\exp(H) \approx 4.38$ for NOVELTY and $\exp(H) \approx 4.31$ for FEASIBILITY, indicating that learning sharpens the mixture while still retaining broad participation rather than degenerating to a single policy.

4.5 Human Study

We run a lightweight expert human study with two goals: (i) assess whether CDQ-generated questions are useful curiosity starting points under a topic-specific evidence packet, and (ii) validate alignment between the LLM-judge win-rate protocol and expert preferences on idea pairs. To reduce overhead, we pool examples across ideation engines. Full annotator instructions are provided in Appendix A.1.4.

Annotators and assigned data points. We recruit 9 PhD students as volunteer annotators. We sample 15 topics and evaluate both NOVELTY and FEASIBILITY, and each data point is labeled by three

Table 5: Human study results involving 9 domain experts across 15 datapoints. “Yes-rate” indicates the proportion of curiosity-driven questions rated as productive starting points. Agreement is measured as the percentage of topics where at least two out of three experts reached a consensus on the “Yes” judgement for question usefulness (Majority Agreement).

Metric	Novelty	Feasibility	All
Yes-rate (%)	95.6	97.8	96.7
Human CDQ win-rate (%)	86.7	80.0	83.3
Agreement (%)	93.3	86.7	90.0
Cohen’s Kappa (κ)	0.76	0.60	0.81

annotators. We use the same assignment for data points for both tasks.

Task 1: Question usefulness. Annotators see the evidence packet and CDQ’s top-10 questions $Q_t^{(y)}$, then choose from {Yes, No, CannotDecide} for whether the question is a useful curiosity starting point for ideation under the target y . We report the Yes-rate and agreement statistics in Table 5.

Task 2: Human validation of LLM-judge winners. For each item (t, y) , we sample an ideation engine e uniformly from {single-pass, RAG, self-reflection} and generate an anonymized idea pair $(I_{t,e,y}^+, I_{t,e,y}^0)$ under the same evidence packet. Annotators choose a winner in the pair of ideas. After recording the independent choice, we reveal the LLM-judge winner and collect an explicit agree/disagree label to measure topic-level alignment. We report: (i) human win-rate of CDQ ideas (majority vote over three), (ii) LLM-human agreement on the majority winner and Cohen’s κ [9], and (iii) individual agree-rate in Table 5.

Task 3: Absolute-quality ratings from human experts. To complement relative pairwise preferences, the same experts also assign absolute quality scores to the CDQ and non-CDQ ideas. They are asked to rate each idea separately for NOVELTY and FEASIBILITY on a 1–10 scale, where 1 denotes the lowest quality, and 10 denotes the highest quality.

Results. Table 5 summarizes the results of the human study conducted with nine domain experts at 15 data points, covering both the usefulness of the question and the preference for the idea. Across both tasks, experts find CDQ questions useful as starting points for curiosity, noting that the generated inquiries effectively target critical knowledge gaps. Furthermore, human preferences in idea pairs align strongly with LLM judge winners, which is reflected in the high agreement and Cohen’s κ scores in the NOVELTY and FEASIBILITY clusters. This substantial alignment between expert consensus and model judgment supports the reliability of our win-rate evaluation protocol and suggests that our automated evaluation accurately captures research quality as perceived by human experts. For absolute-quality ratings from human experts, on average, CDQ increases NOVELTY ratings from 4 to 8 and FEASIBILITY ratings from 3 to 7. While absolute scores can vary across annotators and are therefore used only as a complementary sanity check, these results suggest that CDQ improves not only relative preference but also the absolute perceived quality of generated ideas.

Table 6: Estimated per-topic cost (USD) for CDQ question generation. Token counts are averaged over topics; prices are listed per 1M tokens. We assume no prompt caching, hence these are conservative upper bounds.

Model	T_{in}	T_{out}	$(c_{\text{in}}, c_{\text{out}})$	Cost/topic
GPT-4o	2.23M	50.0K	(\$2.50, \$10.00)	\$6.07
Qwen	2.15M	66.5K	(\$0.09, \$1.10)	\$0.27
Gemini	1.43M	24.5K	(\$0.30, \$2.50)	\$0.49
MiniMax	1.68M	28.8K	(\$0.254, \$1.02)	\$0.46

Brief case studies. The effect of CDQ is visible in individual examples according to human experts. For a malaria diagnostics topic, baseline engines tended to produce broad ideas such as combining portable spectroscopy with AI for field detection. In contrast, CDQ questions directed the engine toward more discriminative technical issues, such as low-parasitemia sensitivity, the limitations of matchbox-sized NIR devices, and whether hemozoin Raman signatures could address overlapping spectral signals. The resulting ideas became more specific, focusing on hybrid spectral designs, calibration choices, and concrete evaluation axes rather than generic AI-assisted diagnosis. Similarly, for a protein-disorder prediction topic, CDQ questions pointed to executable evidence paths such as post-translational modification and linear-motif resources, helping the engine move away from speculative high-level proposals toward data-grounded feature integration. These examples are consistent with the aggregate findings: CDQ improves ideation by changing the problem scaffold available to the generator, encouraging it to reason about discriminative evidence and implementation levers instead of expanding the topic generically. We include additional full examples and raw generations in the Appendix.

4.6 Cost for Deployment

We estimate the API cost of running the CDQ *question-generation* stage for one topic in Table 6. Let T_{in} and T_{out} denote the total input and output tokens (consumed across all CDQ rounds and all calls within the question-generation stage) averaged per topic. For a model with per-1M-token prices $(c_{\text{in}}, c_{\text{out}})$ for input and output tokens, respectively, the expected cost is

$$\text{Cost/topic} \approx \frac{T_{\text{in}}}{10^6} c_{\text{in}} + \frac{T_{\text{out}}}{10^6} c_{\text{out}}. \quad (13)$$

Despite operating over multiple rounds, CDQ is *output-light*: the total output per topic is only 25–66K tokens. When paired with efficient serving endpoints (e.g., Qwen3, Gemini Flash, MiniMax-M2), the end-to-end cost of producing a CDQ question set is under \$0.50 per topic, making large-scale deployment practical (e.g., $\leq$ \$50 for 100 topics). Moreover, some providers offer discounted cached-input rates for repeated prompt prefixes, which can further reduce the effective cost in practice.

The empirical patterns in previous sections and cost for deployment analysis suggest two practical guidelines for using CDQ. First, CDQ is most useful when the downstream engine can turn questions into concrete information-seeking actions. This is consistent with the especially strong FEASIBILITY gains observed for RAG, where CDQ questions can direct retrieval toward implementation

constraints, measurable evidence, and evaluation choices. Second, CDQ should be viewed as a front-end inquiry planner rather than a replacement for the ideation engine. Across single-pass, RAG, and self-reflection pipelines, the same top- K question set provides a shared problem scaffold that encourages the generator to focus on corpus-relative gaps and decision-relevant levers instead of falling back to generic future-work directions.

5 Limitations and Ethical Considerations

Limitations. CDQ operates on a compact evidence packet built from paper titles/abstracts, so it may miss details only available in full texts or supplementary materials, and its question quality depends on retriever coverage and black-box LLM behavior. Our primary evaluation relies on LLM-judged pairwise preferences; despite randomized ordering and repeated voting, any residual noise or bias mainly reflects the judge model’s pre-training data and inductive priors rather than CDQ itself.

Ethical considerations. Our dataset is derived from publicly available scientific metadata (titles/abstracts) and does not involve personal or sensitive user data; our human study used informed consent. As with other ideation tools, CDQ could be misused to facilitate low-effort paper writing, plagiarism, or misleading scientific claims; therefore, we position it as an assistive module that requires human oversight, proper citation/attribution, and domain-expert verification before any real-world scientific use.

6 GenAI Disclosure

We used a generative AI assistant solely for language editing (wording and grammar) of the manuscript. The tool was *not* used to generate or modify any research ideas, methods, experimental designs, data, results, analyses, or conclusions reported in this paper. All technical content and scientific claims were authored and verified by the authors.

7 Conclusion

In this work, we proposed **Curiosity-Driven Questioning (CDQ)**, a training-free, black-box method that learns a compact set of curiosity-driven questions from a topic-specific literature snapshot. CDQ measures the quality of the question relative to the provided corpus using three complementary indices: GAPEXPOSURE to avoid asking questions about the existing knowledge in the provided corpus, DISAGRDIVERSITY to prioritize questions that open competing hypotheses under the same evidence, and CLARITYGAIN to ensure the relevance of the decision by reducing the uncertainty of novelty/feasibility. An inference-time mirror-ascent update over a small library of meta-policies enables CDQ to concentrate on questions that yield higher utility within a few rounds. Our promising results show that CDQ’s enhancement is engine-agnostic and can be plugged into all kinds of LLM research ideation pipelines.

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A Appendix

The appendix is organized as follows:

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A.1 Implementation Details

A.1.1 Code and Dataset Availability. All the code and the dataset we use for our work can be found at this link:

<https://anonymous.4open.science/r/anonymous-cdq-8E5F/>

A.1.2 Details on CDQ pipeline implementations. In this subsection, we list more details about CDQ pipeline implementations, including dataset-preprocessing, algorithm overview, hyperparameter configuration, and the prompt templates we use to profile each component in the CDQ pipeline.

Dataset pre-processing for evidence packet. Directly providing $\mathcal{P}_t$ to an LLM for each query may require substantial resources. We therefore expose $\mathcal{P}_t$ through a lightweight retriever R that returns a small evidence packet $\mathcal{E}(q) = \{s_1, \dots, s_m\}$:

$$\mathcal{E} := R(q, \mathcal{P}_t) = \{s_1, \dots, s_m\}. \quad (14)$$

We implement R with a BM25-style bag-of-words retriever over each document’s concatenated title+abstract [34]. For each query q , we retrieve the top k_{docs} documents and extract up to k_{sent} support sentences per document by token-overlap scoring. This produces a short, citation-friendly evidence packet $\mathcal{E}(q)$ that is used by all agents in our CDQ pipeline.

Algorithm overview. Algorithm 1 illustrates the overview of our CDQ pipeline.

Proxy-gated computation for GAPEXPOSURE. Many candidates are obviously “gappy” and do not require repeated sampling. We thus compute a cheap proxy from the judge:

$$\begin{aligned} (\tilde{\tau}, \tilde{r}) &\sim J(\cdot \mid q, \mathcal{E}(q)), \\ \tilde{\tau} &\in \{\text{Answerable}, \text{Partial}, \text{Unknown}\}, \\ \tilde{r} &\in \{\text{Entails}, \text{Contradicts}, \text{Unrelated}\}. \end{aligned}$$

We map these to a scalar score

$$s_{\text{proxy}}(q) = \frac{1}{2} \left(s_{\tau}(\tilde{\tau}) + s_r(\tilde{r}) \right), \quad (15)$$

Algorithm 1 CDQ: inference-time curiosity optimization for question generation

- 1: **Input:** Topic corpus $\mathcal{P}_t$, target y , policies Π , rounds $\mathcal{L}$, candidates/round L , survivors r_{surv} .
- 2: Build seed evidence packet $\mathcal{E}_0$ from $\mathcal{P}_t$ and cache baseline entropy H'_y (Eq. 5).
- 3: Initialize mixture $w \leftarrow 1/|\Pi|$.
- 4: **for** $\ell = 1$ **to** $\mathcal{L}$ **do**
- 5: Sample L candidates from mixture w (at least one per $\pi \in \Pi$).
- 6: **for each** candidate question q **do**
- 7: Retrieve evidence packet $\mathcal{E}(q) \leftarrow R(q, \mathcal{P}_t)$.
- 8: Compute $\text{Gap}(q)$ with proxy-gating; compute $\text{Disagr}(q)$.
- 9: **end for**
- 10: Select survivors $\mathcal{S}^{(\ell)}$: top r_{surv} by $\alpha \text{Gap}(q) + \beta \text{Disagr}(q)$.
- 11: **for each** $q \in \mathcal{S}^{(\ell)}$ **do**
- 12: Sample idea drafts and compute $\text{Clarity}_y(q)$.
- 13: Compute $U_y(q)$ (Eq. 8).
- 14: **end for**
- 15: Update mixture w via Eq. 10 using per-policy median utilities.
- 16: Early-stop if round median utility does not improve.
- 17: **end for**
- 18: **Output:** Top- K questions by $U_y(q)$ across rounds.

Table 7: Default CDQ configuration (per topic).

Hyperparameter	Value
Rounds $\mathcal{L}$ (cap)	5
Candidates/round L	30
Survivors r_{surv}	15
Gap samples S_g	3
Disagreement samples S_d	5
Clarity idea drafts S_c	5
Baseline panel repeats N_{baseline}	2
Post-question repeats $N_{\text{post-question}}$	3
Weights (α, β, γ)	(0.2, 0.3, 0.5)
EG step size η	1.0
BM25 (k_1, b)	(1.6, 0.75)
Top docs k_{docs}	8
Support sents/doc k_{sent}	6

with

$$\begin{aligned} s_{\tau}(\text{Answerable}) &= 0, & s_{\tau}(\text{Partial}) &= 0.5, \\ s_{\tau}(\text{Unknown}) &= 1, & s_r(\text{Entails}) &= 0, \\ s_r(\text{Unrelated}) &= 0.7, & s_r(\text{Contradicts}) &= 1. \end{aligned} \quad (16)$$

We calibrate an online threshold τ_{α} after a small number of fully-scored questions so that “skipping” full estimation has a low empirical false-skip rate (controlled by α). When $s_{\text{proxy}}(q) \geq \tau_{\alpha}$, we set $\text{Gap}(q) \leftarrow 1$; otherwise, we compute Eq. (2) using S_g samples.

Hyperparameter configuration. Table 7 lists the default hyperparameter configuration for our CDQ pipeline.

LLM agent roles and prompt templates. CDQ instantiates three roles from the same base LLM with role-specific system prompts, and *all* calls are conditioned on the retrieved evidence packet $\mathcal{E}(q)$ for corpus grounding. The **Generator** (*G*) proposes candidate questions under a sampled meta-policy; the **Evaluator** (*E*) returns short evidence-only answers with an answerability tag (Answerable/Partial/Unknown); and the **Judge** (*J*) issues one-token verdicts used by proxy gating and pairwise answer-relation clustering. The exact prompt templates for these agents are shown in Fig. 3.

Lever panel Z_y and prompt templates. To operationalize decision-relevant uncertainty for target $y \in \{\text{NOVELTY}, \text{FEASIBILITY}\}$, we define an interpretable lever panel $Z_y = (Z_{y,1}, \dots, Z_{y,8})$. Each lever corresponds to a concrete aspect along which a question or idea may reduce uncertainty for novelty or feasibility (e.g., boundary regimes, evaluation protocols, data availability, or confound control). For CLARITYGAIN, CDQ prompts a dedicated clarity evaluator to propose a short idea draft and explicitly list the activated levers, after which the judge labels lever states under both corpus-only and post-question contexts. The exact prompt templates are provided in Fig. 4 and Fig. 5.

Question meta-policies and prompt template. CDQ employs a small library of human-inspired question meta-policies $\Pi = \{\pi_1, \dots, \pi_5\}$, each corresponding to a distinct inquiry operator (e.g., information-gap seeking, analogy transfer, counterfactual stress-testing, failure analysis, and constraint-aware evaluation). In implementation, each meta-policy π_i is realized as a natural-language instruction injected into the generator prompt, steering question generation toward complementary curiosity modes. The policy-conditioned prompt template and the instruction strings are shown in Fig. 6.

A.1.3 Baseline ideation engine details.

Single-pass generation engine. This engine serves as our baseline end-to-end generation pipeline, simulating a direct inference scenario.

- **Baseline Configuration:** The model generates a research idea directly from the topic corpus $\mathcal{P}_t$. This is implemented using the template shown in Figure 7, where the model is tasked with reasoning over themes and gaps implied by the background knowledge alone.
- **CDQ Integration:** To evaluate the model's ability to internalize strategic inquiry, we modify the prompt to include the learned questions Q_t . As shown in Figure 8, the model must explicitly address these guiding questions while synthesizing the idea, testing its capacity for latent reasoning in a zero-shot manner.

Retrieval-augmented ideation (RAG) engine. The RAG engine enhances discovery by incorporating external knowledge beyond the provided corpus via a dual-stage retrieval and synthesis process.

- **Retrieval Phase:** The system first fetches scholarly evidence relevant to the topic via web search. In the baseline setting (Figure 9), the search is general. When CDQ is active (Figure 10), the optimized question serves as the specific search query to retrieve targeted external context.

- **Synthesis Phase:** The final idea is synthesized by conditioning on the corpus, the optimization target, and the retrieved context. This is executed via the template in Figure 11. Notably, because the retrieved evidence is already conditioned on the question, we omit the explicit question from this final prompt to ensure conciseness and avoid semantic redundancy.

Self-reflection based ideation engine. To simulate the rigorous nature of academic discourse, we implement a multi-agent feedback loop centered on a self-reflection mechanism involving two roles: a **Researcher** and a **Reviewer**.

- **Review Phase:** The Reviewer scrutinizes the proposal based on the optimization target. The baseline critique (Figure 12) focuses on internal coherence, while the CDQ-enhanced version (Figure 13) requires the Reviewer to evaluate the idea's alignment with the motivating questions.
- **Refinement Phase:** The Researcher then improves the proposal by integrating the critique. The Researcher utilizes either the standard refinement template (Figure 14) or the question-aware template (Figure 15), depending on the experimental setting. We set a fixed budget of three iterations, treating the final refined draft as the definitive engine output.

A.1.4 Instructions for human study. This subsection provides the full documentation and standardized rubrics provided to the expert annotators. These instructions were displayed via the annotation interface during the two evaluation tasks described in Section 4.6.

Task 1: Question Usefulness Assessment. The following instructions were provided to experts to evaluate whether CDQ-optimized questions constitute productive starting points for inquiry: You will be given a topic evidence packet (titles/abstract snippets) and a set of research questions. Judge whether the question set is a good curiosity starting point for developing a strong research idea for the specified target:

- **Novelty:** Questions should help uncover non-obvious gaps, contradictions, regime shifts, or new evaluation/measurement opportunities relative to the packet. Avoid rating questions as "novel" if they are already directly answered in the packet or are generic.
- **Feasibility:** Questions should clarify practical levers (data/evidence availability, measurable outcomes, procedures, constraints, confounds) that determine whether a project can realistically be executed and evaluated.

Constraints: (1) Use only the evidence packet; do not rely on outside knowledge. (2) Ignore writing style; focus on whether the questions would genuinely guide productive inquiry.

Decision: Choose: Yes (useful), No (not useful), or CannotDecide (insufficient information / too ambiguous).

Task 2: Idea Pair + Alignment Assessment. The following protocol governed the side-by-side comparison of generated ideas and the subsequent alignment check with the LLM judge:

You will see a topic evidence packet and two candidate research ideas (A and B). Select the better idea for the specified target:

CDQ agent prompt templates (G, E, and J)

Generator (G) — System. You are the Generator (G) inside the CDQ curiosity optimizer. Your task is to craft concrete, research-level questions using ONLY the provided corpus snippets. Each question must reference cited evidence, highlight uncertainties, and stay grounded in the corpus. Return JSON with a list under the key questions. Never output prose, code fences, or explanations.

Generator (G) — User. Target dimension: {target_dimension}.
 Topic summary: {topic_summary}.
 Policy instruction: {policy_instruction}.
 Corpus snippets (ground truth context, cite them explicitly):
 - {snippet_1}
 - {snippet_2}
 :
 Produce {num_questions} high-leverage questions. Respond ONLY with JSON: {"questions": ["Question 1?", "Question 2?", "Question 3?"]}.
Evaluator (E) — System. You are the Evaluator (E). You must answer questions ONLY using the provided corpus snippets. If the snippets lack information, say Unknown. Return JSON with keys answer (<=80 tokens) and tag in {Answerable, Partial, Unknown}.
Evaluator (E) — User. Question: {question}.
 Corpus snippets:
 - {snippet_1}
 - {snippet_2}
 :
 Respond as JSON: {"answer": "...", "tag": "Answerable"}.

Judge (J) — System. You are the Judge (J). You issue one-token verdicts for tags, entailment, pairwise relations, and attribute panels. You never hallucinate.

Judge (J) — User (pairwise relation). A: {answer_A}
 B: {answer_B}
 Return JSON: {"relation": "Contradicts"}.

Figure 3: Prompt templates for the three internal CDQ agents.

- **Novelty:** Prefer ideas that are genuinely new relative to the packet (not a paraphrase), and that articulate a clear differentiating claim or angle.
- **Feasibility:** Prefer ideas with a realistic plan (data/evidence path, method, evaluation), and awareness of constraints/-confounds.

Constraints: (1) Use only the evidence packet; do not rely on outside knowledge. (2) Ignore writing style; focus on substance and grounding.

Decision Protocol: You must choose **A** or **B**. After your submission, the LLM-judge decision will be revealed, and you will be asked to indicate whether you Agree or Disagree with its verdict to record individual-level alignment.

A.2 Theoretical Analysis

This section provides supporting theory for the three curiosity indices and the inference-time mirror-ascent update in Section 3. All results are based on finite samples unless stated otherwise.

Probabilistic view of black-box LLM components. Fix a corpus $\mathcal{P}$ and target y . For a question q , the evaluator induces a conditional distribution over corpus-grounded answers $A \sim E(\cdot \mid \mathcal{P}, q)$ and

answerability tags $\tau \in \{\text{Answerable}, \text{Partial}, \text{Unknown}\}$. The judge induces conditional distributions over one-token relational and attribute labels. All probabilities below are taken with respect to the internal randomness of these black-box calls (sampling, decoding stochasticity), conditional on $(\mathcal{P}, q)$.

Notation. Throughout this theory section, we fix a topic and write $\mathcal{P} := \mathcal{P}_t$ for brevity.

A.2.1 Distribution-free risk control for the GAPEXPOSURE proxy gate. Recall Eq. (2). For a fixed question q , define the *true* gap level

$$g(q) := \Pr(\tau = \text{Unknown} \mid \mathcal{P}, q) + \frac{1}{2} \Pr(\tau = \text{Partial} \mid \mathcal{P}, q) \in [0, 1], \quad (17)$$

and the empirical estimate $\text{Gap}(q)$ computed from S_g i.i.d. evaluator calls. Let $s_{\text{proxy}}(q) \in [0, 1]$ denote the cheap proxy computed from judge verdicts (Section 3).

Skip-safe region and false skips. Fix a user-chosen skip-safe threshold $g_0 \in (0, 1)$ (e.g., $g_0 = 0.75$). Define $Y(q) = \{g(q) \geq g_0\}$. A *false skip* occurs when $Y(q) = 0$ but the gate skips full estimation based on the proxy.

Lever panels (Novelty and Feasibility)

Novelty levers.

N1 Mechanism/abstraction transfer: proposes adapting a mechanism, model, or method from one domain/task to a new domain/task in a non-trivial way, with some argument for why/how the transfer might work.

N2 Boundary/regime/counterfactual: explores edge cases, new regimes, failure modes, or counterfactual worlds where existing theories/methods are untested, expected to break, or give conflicting predictions.

N3 Constructive tension & theory-challenge: explicitly identifies contradictions, gaps, or competing explanations in prior work and proposes a way to resolve, unify, or decisively test them.

N4 Evaluation/measurement innovation: introduces new metrics, benchmarks, tasks, protocols, or data-collection schemes that would change how the community evaluates this phenomenon or class of methods.

N5 Generative reach & synthesis: expands the solution/idea space (e.g., new decompositions, modular combinations, or integrative frameworks) rather than proposing a single, isolated point solution.

N6 Data/observation novelty: proposes collecting or leveraging qualitatively new kinds of data or observations (new modality, population, environment, scale, or instrument) that enable questions previously impossible or impractical.

N7 Problem framing / question novelty: formulates a genuinely new or substantially reframed research question, objective, or hypothesis rather than a small variant of standard tasks or benchmarks.

N8 Domain/context expansion: moves an established idea or tool into a substantially new scientific, application, or societal domain where it has not been systematically explored, with a plausible argument that this opens a new line of work there.

Feasibility levers.

F1 Measurable outcomes: clearly defined success criteria, hypotheses, or evaluation metrics are stated so that progress and success can be judged objectively.

F2 Evidence/data path: there is a concrete plan for obtaining the necessary data or evidence (existing datasets, experiments, observations, or surveys), including at least a rough sense of availability or sample size.

F3 Procedure / design sketch: outlines key steps, experimental design, or algorithmic pipeline (even at a high level), making it clear how the idea would actually be executed.

F4 Resource / constraint fit: explicitly considers compute, equipment, data access, ethics/privacy/safety, and time, and argues that the project is plausible under typical lab/organization constraints.

F5 Validity / confound control: anticipates main threats to validity (confounds, biases, baselines, robustness issues) and proposes controls, comparisons, ablations, or analysis strategies to mitigate them.

F6 Skill / expertise alignment: the required skills, infrastructure, or collaborations are realistic for a strong PhD student or typical research team; the writeup does not quietly assume capabilities that are effectively out-of-reach (e.g., “just build GPT-7 from scratch”).

F7 Timeline & decomposition: decomposes the project into stages or milestones (minimal viable experiment, main study, extensions) and is appropriately scoped for a realistic research timescale (e.g., 6–36 months).

F8 Risk management & fallback paths: identifies major technical or experimental risks and proposes fallback strategies, simplifications, or alternative paths that would still yield useful knowledge if the main plan fails.

Figure 4: Lever panels used for novelty and feasibility attribution.

Because the threshold is data-dependent, a reviewer may object to selecting $\hat{\tau}_\alpha$ using only an empirical rate constraint. We therefore state a conformal-style gate that is *distribution-free* under exchangeability, matching the spirit of RCPS [4].

THEOREM A.1 (CONFORMAL FALSE-SKIP CONTROL FOR PROXY GATING). Assume the calibration questions $\{q_i\}_{i=1}^n$ and a future test question q_{n+1} are exchangeable under the candidate-question distribution induced by the generator within a fixed round. For each calibration item, compute $(s_{\text{proxy}}(q_i), Y(q_i))$. Let $\mathcal{I}_0 = \{i \in [n] : Y(q_i) = 0\}$ and $n_0 = |\mathcal{I}_0|$. Define the conformal p-value

$$p_{\text{proxy}}(q) := \frac{1 + \sum_{i \in \mathcal{I}_0} \mathbb{1}\{s_{\text{proxy}}(q_i) \geq s_{\text{proxy}}(q)\}}{n_0 + 1}. \quad (18)$$

Gate rule: skip full $\text{Gap}(q)$ estimation only if $p_{\text{proxy}}(q) \leq \alpha$. Then the marginal false-skip probability is controlled:

$$\Pr(Y(q_{n+1}) = 0 \wedge p_{\text{proxy}}(q_{n+1}) \leq \alpha) \leq \alpha. \quad (19)$$

PROOF. Condition on the event $Y(q_{n+1}) = 0$. Under exchangeability, the multiset $\{s_{\text{proxy}}(q_i) : i \in \mathcal{I}_0\} \cup \{s_{\text{proxy}}(q_{n+1})\}$ is exchangeable. Hence the rank of $s_{\text{proxy}}(q_{n+1})$ among these $n_0 + 1$ values is uniform, implying $\Pr(p_{\text{proxy}}(q_{n+1}) \leq \alpha \mid Y(q_{n+1}) = 0) \leq \alpha$. Multiplying by $\Pr(Y(q_{n+1}) = 0) \leq 1$ yields the stated marginal bound. $\square$

[Threshold form] The p-value rule (18) is equivalent to a quantile threshold: let $s_{(1)}^{(0)} \leq \dots \leq s_{(n_0)}^{(0)}$ be the sorted proxy scores among unsafe calibration items $Y = 0$. Setting $\hat{\tau} = s_{(\lceil (1-\alpha)(n_0+1) \rceil)}^{(0)}$ and skipping only when $s_{\text{proxy}}(q) > \hat{\tau}$ yields the same guarantee.

LEMMA A.2 (CONCENTRATION OF THE EMPIRICAL $\text{Gap}(q)$). Let $T_s \in \{0, \frac{1}{2}, 1\}$ be the numeric encoding of the s -th tag draw (Answerable $\mapsto 0$, Partial $\mapsto \frac{1}{2}$, Unknown $\mapsto 1$). Then $\text{Gap}(q) = \frac{1}{S_g} \sum_{s=1}^{S_g} T_s$ and $\mathbb{E}[T_s] = g(q)$. For any $\delta \in (0, 1)$,

$$\Pr\left(|\text{Gap}(q) - g(q)| \geq \sqrt{\frac{\log(2/\delta)}{2S_g}}\right) \leq \delta. \quad (20)$$

Clarity evaluator prompt templates

Clarity Evaluator — System. Answer the question using the provided corpus snippets. Propose a concrete research idea or experiment and explicitly map how it activates Novelty/Feasibility levers.

User. Question: {question}.

Reference snippets:

- {snippet_1}

- {snippet_2}

:

Attribute levers for {target_dimension}:

{N1: ..., N2: ..., or F1: ..., F2: ...}.

Respond as JSON: {"idea": "...", "grounding": "Corpus|Parametric|Mixed", "levers": ["N1", "N4"]}.

Judge — User (lever labeling). Context: CORPUS_SNIPPETS, QUESTION, IDEA. Assess for each lever whether the idea reduces uncertainty relative to the corpus. Return JSON mapping each lever to {Yes, No, Uncertain}.

Figure 5: Prompt templates for CLARITYGAIN.

Meta-policy instructions and policy-conditioned generator prompt.

info-gap-hunter: Identify concrete knowledge gaps or contradictions (missing evidence, unresolved claims, incompatible findings). Ask questions about what is unknown or inconsistent to explore why these gaps arise, what evidence would decisively resolve them, and how resolution would change interpretation or practice.

analogy-bridger: Map a structurally similar problem from another field (source->target). Ask questions about the transferred mechanism to explore whether, when, and why it should hold, where it should fail, and what observations would distinguish successful transfer from failure.

counterfactual-boundary: Alter a key mechanism, assumption, or regime (invert, remove, or push to an extreme). Ask questions about the resulting consequences to explore which predictions diverge, how to discriminate between the original and altered accounts, and which interventions or observations reveal the difference.

failure-to-question: Start from an observed failure mode or limitation (e.g., optimization instability, data shift, brittleness, safety, calibration). Ask questions about likely root causes to explore minimal isolating interventions and feasible mitigations.

constraint-eval-reframer: Reframe the problem under real constraints (compute, data, latency, privacy, safety) and the need for sound evaluation (proxies, constructs, stress tests). Ask questions about key trade-offs and measurement choices to explore robust designs under limits, construct validity, sensitivity/specificity to the intended phenomenon, and predictive utility for downstream outcomes.

Figure 6: Meta-policy instructions and their injection into the generator prompt.

PROOF. $T_s \in [0, 1]$, so Hoeffding's inequality applies to the sample mean. $\square$

A.2.2 Concentration and robustness for DISAGRDIVERSITY. [Latent semantic answer types] For a fixed q , there exist K^* latent semantic answer types with probabilities $p^* = (p_1^*, \dots, p_{K^*}^*)$. Each evaluator sample first draws a type $k \sim p^*$ and then generates a surface answer a consistent with that type.

Perfect-oracle baseline. Assume first an oracle that clusters answers perfectly by type, yielding empirical frequencies $\hat{p}_k = N_k/S_d$

from S_d i.i.d. answers. Let $H(p) = -\sum_k p_k \log p_k$ denote Shannon entropy (nats).

THEOREM A.3 (MILLER–MADOW: ESTIMATION ERROR). Let $\hat{H}_{MM} = H(\hat{p}) + \frac{\hat{K}-1}{2S_d}$ be the Miller–Madow estimator computed from the induced clusters, where $\hat{K}$ is the number of observed clusters. Under Assumption A.2.2, for any $\delta \in (0, 1)$,

Single-pass generation engine without question prompt template.

You are a top-tier researcher. You are tasked with creating a research idea given some background knowledge derived from existing papers.

Background knowledge: {background_knowledge}

Using the background knowledge provided, reason over the key themes, limitations, and open gaps implied by the existing work, and generate a research idea that demonstrates strong {quality_indicator}.

The idea should be well-motivated by the background knowledge and reflect a coherent research direction.

Please avoid copying ideas directly; instead, synthesize and extend the insights to inspire a new research idea.

Return your idea in the following format:

Title: [A brief, focused title]

Problem: [The core issue or gap being addressed]

Objective: [The main goal or research question]

Hypothesis: [The hypothesis being tested or explored]

Method: [The approach or methodology]

Expected Impact/Findings: [The anticipated outcomes or contributions]

Please only respond with the research idea in the format provided above. Do not respond with anything else.

Figure 7: Prompt template for the single-pass ideation engine without questions.

$$|\widehat{H}_{\text{MM}} - H(p^*)| \leq c \sqrt{\frac{K^* \log(1/\delta)}{S_d}} + \frac{K^* - 1}{2S_d}, \quad (21)$$

with probability $\geq 1 - \delta$.

for a universal constant $c > 0$.

PROOF SKETCH. Combine (i) concentration of multinomial proportions $\hat{p}$ around p^* with a union bound over K^* , and (ii) a continuity/Lipschitz argument for entropy on the simplex away from the boundary. The Miller–Madow correction cancels the leading $O(1/n)$ bias term of the plug-in entropy estimator [32]. $\square$

Robustness to noisy entailment clustering. In practice, clustering uses judge entailment and may incur merges/splits. Let $\tilde{p}$ be the empirical cluster-frequency distribution produced by the judge-based clustering, and let $\hat{p}$ be the oracle frequency distribution (types observed in the sample).

PROPOSITION A.4 (ENTROPY STABILITY UNDER CLUSTERING DISTORTION). Assume $\|\tilde{p} - \hat{p}\|_1 \leq \varepsilon_c \leq \frac{1}{2}$ and both are distributions over at most K clusters. Then

$$|H(\tilde{p}) - H(\hat{p})| \leq \varepsilon_c \log(K - 1) + h(\varepsilon_c), \quad (22)$$

where $h(\varepsilon) = -\varepsilon \log \varepsilon - (1 - \varepsilon) \log(1 - \varepsilon)$ is the binary entropy (nats).

PROOF. This is the (classical) Fannes–Audenaert continuity bound for entropy. $\square$

COROLLARY A.5 (TOTAL ERROR DECOMPOSITION FOR Disagr(q)). Under Assumption A.2.2, with probability at least $1 - \delta$,

$$|\text{Disagr}(q) - H(p^*)| \leq c \sqrt{\frac{K^* \log(1/\delta)}{S_d}} + \frac{K^* - 1}{2S_d} + \varepsilon_c \log(K - 1) + h(\varepsilon_c), \quad (23)$$

where ε_c summarizes the clustering distortion induced by judge noise on that sample.

PROOF. Triangle inequality: $|\text{Disagr}(q) - H(p^*)| \leq |\widehat{H}_{\text{MM}} - H(p^*)| + |H(\tilde{p}) - H(\hat{p})|$, then apply Theorem A.3 and Proposition A.4. $\square$

A.2.3 Decision-theoretic meaning of CLARITYGAIN. Recall Eq. (6). Let Z_y denote the attribute panel random variable for target y (a finite-valued vector), with

$$\pi(z) := \Pr(Z_y = z \mid \mathcal{P}), \quad (24)$$

$$\pi_{q,a}(z) := \Pr(Z_y = z \mid \mathcal{P}, q, a). \quad (25)$$

LEMMA A.6 (MUTUAL-INFORMATION IDENTITY).

$$\begin{aligned} \text{Clarity}_y(q) &= \mathbb{E}_{a \sim E(\cdot \mid \mathcal{P}, q)} \text{KL}(\pi_{q,a} \parallel \pi) \\ &= I(Z_y; A \mid \mathcal{P}, q) \geq 0. \end{aligned} \quad (26)$$

PROOF. This is the standard identity $I(U; V \mid W) = \mathbb{E}_v(P(U \mid v, W) \parallel P(U \mid W)) = H(U \mid W) - H(U \mid V, W)$, applied to $U = Z_y$, $V = A$, $W = (\mathcal{P}, q)$. $\square$

Single-pass generation engine with question Prompt Template.

You are a top-tier researcher. You are tasked with creating a research idea given some background knowledge and guiding questions.

The background knowledge comes from existing papers.

The guiding questions clarify the problem setting, underlying assumptions, and practical considerations that shape the research direction.

Background knowledge: {background_knowledge}

Guiding questions: {questions}

Using both the background knowledge and the guiding questions, reason over key themes, limitations, and open gaps, and generate a research idea that demonstrates strong {quality_indicator}.

The idea should be well-grounded in prior work and coherent with the problem setting implied by the guiding questions.

Please avoid copying ideas directly; instead, synthesize insights from both sources to inspire a new research idea.

Return your idea in the following format:

Title: [A brief, focused title]

Problem: [The core issue or gap being addressed]

Objective: [The main goal or research question]

Hypothesis: [The hypothesis being tested or explored]

Method: [The approach or methodology]

Expected Impact/Findings: [The anticipated outcomes or contributions]

Please only respond with the research idea in the format provided above. Do not respond with anything else.

Figure 8: Prompt template for the single-pass ideation engine with questions.

THEOREM A.7 (EXACT BAYES RISK REDUCTION UNDER LOG LOSS). *Under the logarithmic scoring rule, the Bayes risk of predicting Z_y equals the conditional entropy. Therefore,*

$$\mathbb{E}_a[H(Z_y | \mathcal{P}) - H(Z_y | \mathcal{P}, q, a)] = \text{Clarity}_y(q) \geq 0. \quad (27)$$

PROOF. For categorical Z , the expected log-loss of the Bayes posterior equals $H(Z | \cdot)$ (proper scoring rules; see [12]). Apply this before and after observing A , subtract, and use Lemma A.6. $\square$

THEOREM A.8 (UNIVERSAL VALUE-OF-INFORMATION UPPER BOUND). *Consider any downstream decision problem with action $d \in \mathcal{D}$, state $z \in \text{supp}(Z_y)$, and bounded utility $u(d, z) \in [0, 1]$. Let*

$$V(\pi) := \max_{d \in \mathcal{D}} \mathbb{E}_{z \sim \pi}[u(d, z)]. \quad (28)$$

be the optimal value under posterior π . Then

$$0 \leq \mathbb{E}_a[V(\pi_{q,a}) - V(\pi)] \leq \mathbb{E}_a \text{TV}(\pi_{q,a}, \pi) \leq \sqrt{\frac{1}{2} \text{Clarity}_y(q)}. \quad (29)$$

PROOF. The lower bound is Blackwell monotonicity (a richer signal cannot hurt optimal expected utility). For the upper bound, note that $V(\pi)$ is the pointwise maximum of linear functionals $\pi \mapsto$

$\mathbb{E}_\pi[u(d, Z)]$, each 1-Lipschitz in total variation because $u \in [0, 1]$. Thus $|V(\pi_{q,a}) - V(\pi)| \leq \text{TV}(\pi_{q,a}, \pi)$. Finally, Pinsker's inequality gives $\text{TV}(\pi_{q,a}, \pi) \leq \sqrt{\text{KL}(\pi_{q,a} \| \pi) / 2}$, and Lemma A.6 averages the KL. $\square$

Why per-question CLARITYGAIN is a principled proxy for set-level value. The main algorithm ranks questions individually and outputs a compact top set. The following results formalize when per-question information gain is additive or (more generally) submodular.

[Conditional independence of answers given the latent panel] For a set of questions $Q = \{q_1, \dots, q_K\}$, let $A_{1:K}$ be the corresponding answers. Assume $A_1, \dots, A_K$ are conditionally independent given $(Z_y, \mathcal{P}, Q)$.

PROPOSITION A.9 (ADDITIVITY OF INFORMATION GAIN). *Under Assumption A.2.3,*

$$\begin{aligned} I(Z_y; A_{1:K} | \mathcal{P}, Q) &= \sum_{k=1}^K I(Z_y; A_k | \mathcal{P}, q_k) \\ &= \sum_{k=1}^K \text{Clarity}_y(q_k). \end{aligned} \quad (30)$$

Get retrieval context without question prompt template.

You are an expert AI research assistant. Your primary task is to retrieve and synthesize information.

To complete this task, you must actively perform **web searches** to collect the latest advancements, related concepts, and potential challenges.

Provided Research Papers: {papers_info}

Your Task:

1. Search the web for relevant, up-to-date information that complements (but does not duplicate) the content of the provided research papers.
2. Focus your summary on insights, methods, breakthroughs, or challenges that are absent from or not emphasized in the provided papers.
3. Generate a concise yet comprehensive context highlighting: - Key technologies and frameworks - Recent breakthroughs and real-world applications - Open problems and common obstacles

Important: Do not include any URLs, links, or references to specific websites in your output. This synthesized context will later be used to brainstorm a new idea, so prioritize information that adds novel, non-overlapping, and helpful perspectives beyond what is already contained in the research papers.

Figure 9: Prompt template for retrieving context without questions.

Get retrieval context with question prompt template.

You are an expert AI research assistant. Your primary task is to retrieve and synthesize information.

To complete this task, you must actively perform **web searches** to collect the latest advancements, related concepts, and potential challenges.

Provided Research Papers: {papers_info}

Questions: {question}

Your Task:

1. Search the web for relevant, up-to-date information that complements (but does not duplicate) the content of the provided research papers.
2. Focus your summary on insights, methods, breakthroughs, or challenges that are **absent from or not emphasized in the provided papers**, but are useful in addressing the given questions.
3. Generate a concise yet comprehensive context highlighting: - Key technologies and frameworks - Recent breakthroughs and real-world applications - Open problems and common obstacles

This synthesized context will later be used to brainstorm a new idea, so prioritize information that adds novel, non-overlapping, and helpful perspectives beyond what is already contained in the research papers.

Figure 10: Prompt template for retrieving context with questions.

PROOF. Apply the chain rule of mutual information: $I(Z; A_{1:K} | W) = \sum_{k=1}^K I(Z; A_k | W, A_{<k})$. Conditional independence implies $I(Z; A_k | \mathcal{P}, Q, A_{<k}) = I(Z; A_k | \mathcal{P}, q_k)$, yielding the sum. $\square$

THEOREM A.10 (MONOTONE SUBMODULARITY OF INFORMATION GAIN). *Under Assumption A.2.3, the set function $F(Q) := I(Z_y; A_Q | \mathcal{P}, Q)$ is monotone and submodular in Q . Consequently, the greedy algorithm that iteratively adds the question with the largest marginal*

RAG engine prompt template.

You are a top-tier researcher. You are tasked with creating a research idea given some background knowledge.

The background knowledge comes from existing papers, and the questions highlight the research directions:

1. Background knowledge: {background_knowledge}
2. Additional context retrieved from web search: {rag_context}

Using the knowledge provided, reason over them and generate a research idea that demonstrates strong {quality_indicator}. Please avoid copying ideas directly; rather, use the insights to inspire a new research idea.

Return your idea in the following format:

Title: [A brief, focused title]

Problem: [The core issue or gap being addressed]

Objective: [The main goal or research question]

Hypothesis: [The hypothesis being tested or explored]

Method: [The approach or methodology]

Expected Impact/Findings: [The anticipated outcomes or contributions]

Please only respond with the research idea in the format provided above. Do not respond with anything else.

Figure 11: Prompt template for generating ideas using the retrieved context.

Reviewer prompt template without questions.

You are a reviewer evaluating a research proposal.

Your task is to provide constructive criticism on the research idea's {quality_indicator}, helping the proposer improve it as much as possible.

Base your critique strictly on the proposed research idea itself. Use the idea to infer the intended problem setting, underlying assumptions, and evaluation context implied by the proposal.

Assess whether the idea is internally coherent and sufficiently developed under its implied setting, and identify gaps, ambiguities, or unrealistic assumptions that limit the {quality_indicator}.

Explain why these issues matter given what the idea claims to address.

In addition to identifying weaknesses, suggest concrete and actionable improvements (e.g., clarifying assumptions, refining the method, or strengthening the evaluation plan) that would improve the idea's {quality_indicator} within the scope defined by the proposal.

Proposed Research Idea: {research_idea}

Figure 12: Prompt template for the reviewer agent providing feedback on research ideas without questions.

increase in F achieves a $(1 - 1/e)$ -approximation to the optimal size- K set.

PROOF SKETCH. Monotonicity follows because mutual information is nonnegative and adding observations cannot decrease information. Submodularity follows from diminishing returns of information gain under conditional independence (standard “sensor

Reviewer prompt template with questions.

You are a reviewer evaluating a research proposal.

Your task is to provide constructive criticism on the research idea's {quality_indicator}, helping the proposer improve it as much as possible.

Base your critique strictly on the proposed research idea itself. Use the idea to infer the intended problem setting, underlying assumptions, and evaluation context implied by the proposal.

Assess whether the idea is internally coherent and sufficiently developed under its implied setting, and identify gaps, ambiguities, or unrealistic assumptions that limit the {quality_indicator}.

Explain why these issues matter in the context implied by the questions.

In addition to identifying weaknesses, suggest concrete and actionable improvements (e.g., clarifying assumptions, refining the method, or strengthening the evaluation plan) that would make the idea better aligned with the problem setting suggested by the questions and improve its {quality_indicator}.

Motivating Questions: {question}

Proposed Research Idea: {research_idea}

Figure 13: Prompt template for the reviewer agent providing feedback on research ideas with questions.

Researcher prompt template without questions.

You are a top-tier researcher proposing research ideas. Your role is to create a research idea and refine the idea if you receive feedback.

A reviewer will review your research idea based on its {quality_indicator} and give you feedback. You should try your best to improve the idea based on the reviewer's feedback and your expertise, especially paying attention to the idea's {quality_indicator}.

Here is the previous idea: {previous_idea}

Here is the reviewer's feedback: {reviewer_feedback}

Based on the reviewer's feedback regarding the previous research idea's {quality_indicator}, generate a revised and improved research idea using the following format:

Title: [A brief, focused title]

Problem: [The core issue or gap being addressed]

Objective: [The main goal or research question]

Hypothesis: [The hypothesis being tested or explored]

Method: [The approach or methodology]

Expected Impact/Findings: [The anticipated outcomes or contributions]

Please only respond with the improved research idea in the format provided above. Do not respond with anything irrelevant.

Figure 14: Prompt template for the researcher agent generating revised ideas based on reviewer feedback without prior questions.

Researcher prompt template with questions.

You are a top-tier researcher. You are tasked with creating a research idea given some background knowledge and guiding questions.

The background knowledge comes from existing papers. The guiding questions clarify the problem setting, underlying assumptions, and practical considerations that shape the research direction.

Background knowledge: {background_knowledge}

Guiding questions: {questions}

Using both the background knowledge and the guiding questions, reason over key themes, limitations, and open gaps, and generate a research idea that demonstrates strong {quality_indicator}.

The idea should be well-grounded in prior work and coherent with the problem setting implied by the guiding questions.

Please avoid copying ideas directly; instead, synthesize insights from both sources to inspire a new research idea.

Return your idea in the following format:

Title: [A brief, focused title]

Problem: [The core issue or gap being addressed]

Objective: [The main goal or research question]

Hypothesis: [The hypothesis being tested or explored]

Method: [The approach or methodology]

Expected Impact/Findings: [The anticipated outcomes or contributions]

Please only respond with the research idea in the format provided above. Do not respond with anything else.

Figure 15: Prompt template for the researcher agent generating revised ideas based on reviewer feedback with prior questions.

placement” argument). The $(1 - 1/e)$ guarantee follows from the classical Nemhauser–Wolsey–Fisher theorem for monotone submodular maximization under a cardinality constraint. $\square$

[Connection to CDQ’s decoupled utility] When Assumption A.2.3 approximately holds, ranking questions by $\text{Clarity}_y(q)$ (or the full utility $U_y(q)$) approximates greedy maximization of a principled set-level objective $F(Q)$. Violations of conditional independence correspond to redundancy between questions; empirically we mitigate redundancy by selecting a compact set and by diversity-promoting meta-policies.

A.2.4 Two-stage screening: compute–statistical efficiency for CLARITYGAIN. In our implementation (Section 3.3), we compute the cheap indices (GAPEXPOSURE and DISAGR DIVERSITY) for all L candidate questions, cull to r_{surv} survivors using only these cheap signals, and compute CLARITYGAIN only for the survivors. This subsection formalizes when this two-stage design is “safe” in the sense that it does not discard any question whose *total* utility is above a (data-dependent) threshold, under mild concentration assumptions for the cheap estimators.

Stage-1 score and population utility. Let q be a candidate question in a fixed round (with fixed $\mathcal{P}$ and target y). Define the *population* (expectation) versions of the indices:

$$g(q) := \mathbb{E}[\text{Gap}(q)], \quad (31)$$

$$d(q) := \mathbb{E}[\text{Disagr}(q)], \quad (32)$$

$$c_y(q) := \text{Clarity}_y(q). \quad (33)$$

where the expectation is over evaluator/judge randomness for a fixed $(\mathcal{P}, q)$. Define the stage-1 population score

$$V(q) := \alpha g(q) + \beta d(q), \quad (34)$$

and the (unnormalized) population total utility

$$U_y^*(q) := V(q) + \gamma c_y(q). \quad (35)$$

(Equation (35) is the population analogue of the per-round utility used in Eq. (8). Any fixed bounded rescaling (including min–max normalization) preserves the screening argument below by replacing constants with the rescaled range.)

Stage-1 estimator and survivor set. Let $\widehat{g}(q)$ and $\widehat{d}(q)$ be the finite-sample estimates computed from S_g and S_d evaluator samples respectively, and define

$$\widehat{V}(q) := \alpha \widehat{g}(q) + \beta \widehat{d}(q). \quad (36)$$

Let $Q = \{q_1, \dots, q_L\}$ be the candidate set in the round and let $S_r \subset Q$ be the set of r_{surv} survivors with the largest $\widehat{V}(q)$ values (ties broken arbitrarily). Write

$$\widehat{v}_{(r)} := \min_{q \in S_r} \widehat{V}(q), \quad (37)$$

so that every discarded question $q \notin \mathcal{S}_r$ satisfies $\widehat{V}(q) \leq \widehat{v}_{(r_{\text{surv}})}$.

A universal upper bound for CLARITYGAIN.

LEMMA A.11 (CLARITYGAIN IS UNIFORMLY BOUNDED). *Let $\mathcal{Z}_y$ be the (finite) state space of the attribute panel Z_y . Then for every q ,*

$$\begin{aligned} 0 \leq c_y(q) &= I(Z_y; A \mid \mathcal{P}, q) \\ &\leq H(Z_y \mid \mathcal{P}) \leq \log |\mathcal{Z}_y|. \end{aligned} \quad (38)$$

In particular, if the panel has M attributes each labeled in $\{\text{Yes, No, Uncertain}\}$, then $|\mathcal{Z}_y| \leq 3^M$ and $c_y(q) \leq M \log 3$ (nats).

PROOF. Mutual information is nonnegative and upper bounded by the entropy of either variable: $I(Z_y; A \mid \cdot) \leq H(Z_y \mid \cdot) \leq \log |\mathcal{Z}_y|$. The cardinality bound follows from the panel definition. $\square$

Let $C_{\max} := \log |\mathcal{Z}_y|$ denote this bound.

Uniform concentration for the stage-1 score.

LEMMA A.12 (UNIFORM DEVIATION BOUND FOR $\widehat{V}$). *Fix a round with L candidates $\mathcal{Q}$. Assume (i) $\widehat{g}(q)$ is computed as an average of S_g i.i.d. bounded tag encodings in $[0, 1]$ (Lemma A.2), and (ii) $\widehat{d}(q)$ is computed via the Miller–Madow estimator from S_d i.i.d. answers under Assumption A.2.2, with $K^*(q) \leq K_{\max}$ for all $q \in \mathcal{Q}$ in the round. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\begin{aligned} \sup_{q \in \mathcal{Q}} |\widehat{V}(q) - V(q)| &\leq \varepsilon_V(\delta) \\ &:= \alpha \sqrt{\frac{\log(4L/\delta)}{2S_g}} \\ &\quad + \beta \left(c \sqrt{\frac{K_{\max} \log(2L/\delta)}{S_d}} + \frac{K_{\max} - 1}{2S_d} \right). \end{aligned} \quad (39)$$

for the same universal constant c as in Theorem A.3.

PROOF. Apply Lemma A.2 with failure probability $\delta/(2L)$ and take a union bound over L candidates to control $\sup_q |\widehat{g}(q) - g(q)|$. Apply Theorem A.3 with failure probability $\delta/(2L)$ and a union bound to control $\sup_q |\widehat{d}(q) - d(q)|$. Combine with $\widehat{V} - V = \alpha(\widehat{g} - g) + \beta(\widehat{d} - d)$. $\square$

Screening guarantee: no high-utility question is discarded.

THEOREM A.13 (TWO-STAGE SCREENING GUARANTEE). *Let $\mathcal{S}_r$ be the stage-1 survivor set (top r_{surv} by $\widehat{V}$). On the event of Lemma A.12, every discarded question $q \notin \mathcal{S}_{r_{\text{surv}}}$ satisfies the population upper bound*

$$U_y^*(q) \leq \widehat{v}_{(r_{\text{surv}})} + \varepsilon_V(\delta) + \gamma C_{\max}, \quad (40)$$

where $\widehat{v}_{(r_{\text{surv}})}$ is defined in Eq. (37) and $C_{\max}$ in Lemma A.11. Equivalently, with probability at least $1 - \delta$,

$$U_y^*(q) > \widehat{v}_{(r)} + \varepsilon_V(\delta) + \gamma C_{\max} \implies q \in \mathcal{S}_r. \quad (41)$$

PROOF. Fix any $q \notin \mathcal{S}_r$. By definition of $\widehat{v}_{(r_{\text{surv}})}$, $\widehat{V}(q) \leq \widehat{v}_{(r_{\text{surv}})}$. On the event of Lemma A.12, we have $V(q) \leq \widehat{V}(q) + \varepsilon_V(\delta) \leq \widehat{v}_{(r_{\text{surv}})} + \varepsilon_V(\delta)$. Finally, $U_y^*(q) = V(q) + \gamma c_y(q) \leq \widehat{v}_{(r_{\text{surv}})} + \varepsilon_V(\delta) + \gamma C_{\max}$ by Lemma A.11. $\square$

[Interpretation] Theorem A.13 provides a *data-dependent* screening threshold: once $\widehat{v}_{(r)}$ is known, any question whose population utility exceeds $\widehat{v}_{(r)} + \varepsilon_V(\delta) + \gamma C_{\max}$ is guaranteed (with probability $\geq 1 - \delta$) to survive stage 1. Thus, evaluating CLARITYGAIN only on $\mathcal{S}_r$ cannot discard any “clearly high-utility” question.

Optional: a posteriori certificate for top-K containment. If desired, we can turn the screening bound into a simple certificate that the global top-K questions (by population utility) must lie inside $\mathcal{S}_r$.

COROLLARY A.14 (TOP-K CONTAINMENT CERTIFICATE). *Suppose we compute a (possibly noisy) estimate $\widehat{U}(q)$ of $U_y^*(q)$ for every $q \in \mathcal{S}_r$, and let $\widehat{u}_{(K)} := \min\{\widehat{U}(q) : q \text{ is among the top-}K \text{ in } \mathcal{S}_r\}$. If*

$$\widehat{u}_{(K)} > \widehat{v}_{(r_{\text{surv}})} + \varepsilon_V(\delta) + \gamma C_{\max}, \quad (42)$$

then (on the same event as Theorem A.13) no discarded question can enter the global top-K set by population utility U_y^ . In other words, under (42), all globally top-K questions are contained in $\mathcal{S}_r$.*

PROOF. By Theorem A.13, every discarded $q \notin \mathcal{S}_r$ has $U_y^*(q) \leq \widehat{v}_{(r_{\text{surv}})} + \varepsilon_V(\delta) + \gamma C_{\max}$. If $\widehat{u}_{(K)}$ exceeds this bound, then the K -th best survivor (hence also the top K survivors) has utility larger than every discarded candidate, so no discarded candidate can be in the global top K . $\square$

A.2.5 *Multiplicative-weights guarantees for the EG update.* Let $\Pi = \{\pi_1, \dots, \pi_{|\Pi|}\}$ be the meta-policy set ($|\Pi| = 5$ in our implementation). At round ℓ , let $g_i^{(\ell)} \in [0, 1]$ denote the per-policy feedback used in Eq. (10) (in our method, the median utility among survivors from π_i). The update is

$$w_i^{(\ell+1)} \propto w_i^{(\ell)} \exp\{\eta g_i^{(\ell)}\}, \quad w^{(1)} = \mathbf{1}/|\Pi|. \quad (43)$$

THEOREM A.15 (EG SOLVES THE PROXIMAL SUBPROBLEM). *For any round ℓ , the update $w^{(\ell+1)}$ is the unique maximizer of*

$$\max_{w' \in \Delta^{|\Pi|}} \left\{ \langle w', g^{(\ell)} \rangle - \frac{1}{\eta} \text{KL}(w' \| w^{(\ell)}) \right\}, \quad (44)$$

equivalently entropic mirror ascent (multiplicative weights).

PROOF. Standard KKT derivation (identical to the main-body derivation after Eq. (10)). $\square$

THEOREM A.16 (REGRET BOUND (ADVERSARIAL SEQUENCE)). *Assume $g_i^{(\ell)} \in [0, 1]$ for all i, ℓ and $\eta \in (0, 1]$. Then for any fixed policy $i^* \in [|\Pi|]$,*

$$\sum_{r=1}^R \langle w^{(r)}, g^{(r)} \rangle \geq \sum_{r=1}^R g_{i^*}^{(r)} - \frac{\log |\Pi|}{\eta} - \frac{\eta T}{8}. \quad (45)$$

PROOF SKETCH. This is the classical multiplicative-weights analysis using the potential $\Phi^{(\ell)} = \log \sum_i w_i^{(\ell)} \exp(\eta \sum_{s \leq \ell-1} g_i^{(s)})$, the inequality $\log \mathbb{E}[\exp(\eta X)] \leq \eta \mathbb{E}[X] + \eta^2/8$ for $X \in [0, 1]$, and telescoping; see, e.g., [1]. $\square$

COROLLARY A.17 (ROBUSTNESS TO UTILITY ESTIMATION ERROR).
 Suppose the feedback $g_i^{(\ell)}$ is an estimate of a latent utility $\mu_i^{(\ell)} \in [0, 1]$
 with uniform error $|g_i^{(\ell)} - \mu_i^{(\ell)}| \leq \varepsilon_\ell$. Then

$$\sum_{\ell=1}^L \langle w^{(\ell)}, \mu^{(\ell)} \rangle \geq \max_i \sum_{\ell=1}^R \mu_i^{(\ell)} - \frac{\log m}{\eta} - \frac{\eta R}{8} - 2 \sum_{\ell=1}^R \varepsilon_\ell. \quad (46)$$

PROOF. Apply Theorem A.16 to $g^{(\ell)}$, then substitute $\mu^{(\ell)}$ and bound the difference by $\sum_t |\langle w^{(\ell)}, g^{(\ell)} - \mu^{(\ell)} \rangle| \leq \sum_\ell \varepsilon_\ell$ and $\max_i \sum_t (g_i^{(\ell)} - \mu_i^{(\ell)}) \leq \sum_\ell \varepsilon_\ell$. $\square$

A.3 Case Studies

To illustrate the technical evolution facilitated by the Curiosity-Driven Questioning (CDQ) framework, we present two detailed case studies. These examples contrast the outputs of multiple ideation attempts before and after the integration of CDQ.

A.3.1 Case Study I: Advancing Malaria Diagnostics.

Complete Evidence Snippets (Background Context).

- (1) **Snippet 1:** Ultraviolet/Visible and Near-Infrared Dual Spectroscopic Method for Detection and Quantification of Low-Level Malaria Parasitemia in Whole Blood. Abstract: Testing methods that are ultrasensitive to low-level ring-staged parasitemia are urgently required. In this study, we combined electronic UV/Vis and NIR spectroscopy to detect and quantify low-level (1-0.000001%) ring-staged malaria-infected whole blood.
- (2) **Snippet 2:** A Near-Infrared "Matchbox Size" Spectrometer to Detect and Quantify Malaria Parasitemia. Abstract: Inexpensive "matchbox size" NIR spectrophotometer used to detect and quantify malaria infection in vitro from isolated dried red blood cells using a fingerpick volume of blood.
- (3) **Snippet 3:** Performance of a High-Sensitivity Rapid Diagnostic Test for *P. falciparum* Malaria. Abstract: The performance of a new Alere Malaria Ag P.f Ultra Sensitive rapid diagnostic test (uRDT) was compared with the SD Bioline Malaria Ag P.f RDT.
- (4) **Snippet 4:** Morphology-sensitive low wavenumber modes of the malaria pigment hemozoin. Abstract: Resonance Raman spectroscopy investigating hemozoin was selectively enhanced with excitation wavelengths at $\lambda = 633$ nm and $\lambda = 647$ nm.
- (5) **Snippet 5:** Limited Level of Accuracy Provided by Available Rapid Diagnosis Tests for Malaria. Abstract: Accuracy of RDTs is insufficient and could increase the number of incorrect malaria diagnoses.
- (6) **Snippet 6:** Spectrophotometry of Human Hemoglobin in the near Infrared Region from 1000 to 2500 nm. Abstract: At wavelengths longer than 1500 nm, the absorbance and second derivative absorbance spectra of hemoglobin species are similar.
- (7) **Snippet 7:** Parasight F test compared with PCR and microscopy for the diagnosis of *P. falciparum* malaria in travelers.

- (8) **Snippet 8:** Use and limitations of light microscopy for diagnosing malaria at the primary health care level. Abstract: Malaria diagnosis is based on light microscopy and is likely to remain so, while research into other simplified techniques is pursued.

Baseline Ideation (Set A: Ideas without CDQ).

- **Idea 1: Title:** Integrating Mid-Infrared Spectroscopy with AI for Non-Invasive, Field-Ready Malaria Diagnosis and Surveillance. **Problem:** Current malaria diagnostic methods face challenges in balancing sensitivity, ease of use, and field applicability, especially in low-resource settings. Non-invasive methods are underdeveloped and require further validation for human application. **Objective:** To develop and validate a non-invasive, mid-infrared spectroscopy-based diagnostic tool integrated with AI for accurate detection of malaria in humans and monitoring of transmission dynamics in endemic regions. **Hypothesis:** Mid-infrared spectroscopy combined with AI can accurately detect malaria infection at submicroscopic levels in humans non-invasively, and provide reliable surveillance data on parasite transmission dynamics. **Method:** Develop a portable mid-infrared spectroscopic device with AI algorithms capable of analyzing human skin spectral signatures for malaria detection. Conduct clinical trials in endemic regions to validate accuracy and usability. Integrate digital connectivity for real-time data reporting and alignment with existing surveillance frameworks. **Expected Impact/Findings:** This approach is anticipated to provide a non-invasive, highly accurate, and field-ready diagnostic tool for malaria, enhancing early detection and surveillance capabilities in resource-limited settings. The integration with digital systems may improve real-time monitoring and inform public health strategies for malaria control and elimination.
- **Idea 2: Title:** Integrated Multi-Spectral and Microfluidic Diagnostic Platform for Rapid Malaria Detection. **Problem:** Current diagnostic methods for malaria face challenges in balancing sensitivity, portability, and cost-effectiveness for field deployment, particularly in resource-limited settings. **Objective:** To develop a novel diagnostic platform that combines multi-spectral imaging and microfluidic technology for the rapid, accurate, and cost-effective detection of malaria at low parasitemia levels. **Hypothesis:** An integrated platform utilizing multi-spectral imaging to analyze blood samples combined with microfluidic immunoassays can provide high sensitivity and specificity in malaria detection, while remaining portable and cost-effective for field use. **Method:** The research will involve designing a compact device that incorporates multi-spectral imaging to capture detailed blood sample spectra and microfluidic channels to enable rapid analysis of malaria biomarkers such as PfHRP2. This device will leverage AI algorithms to enhance detection accuracy and will be tested on clinical samples from malaria-endemic regions to evaluate its performance against existing diagnostic methods. **Expected Impact/Findings:** This research is expected to yield a

highly sensitive and specific diagnostic tool that is portable and affordable for use in resource-limited settings. The platform could significantly improve malaria surveillance and management, contributing to global efforts in malaria elimination by enabling early and accurate detection of infections.

- **Idea 3: Title:** Integrating Non-Invasive Near-Infrared Spectroscopy with AI for Malaria Detection in Pregnant Women. **Problem:** Current malaria detection methods, while improving, often face challenges in non-invasive application, particularly in vulnerable populations such as pregnant women, where blood collection is risky or culturally sensitive. **Objective:** To develop and validate a non-invasive, near-infrared spectroscopy (NIR) coupled with AI algorithms for accurate detection of low-level malaria parasitemia in pregnant women. **Hypothesis:** Non-invasive NIR spectroscopy, when enhanced with AI-based pattern recognition, can detect malaria parasitemia at clinically relevant levels in pregnant women with high sensitivity and specificity, comparable to invasive diagnostic methods. **Method:** Utilize a portable NIR device to scan skin surfaces (e.g., hands, feet) of pregnant women in endemic areas. Implement machine learning algorithms to analyze spectral data for malaria biomarkers. Validate the system's performance against standard blood-based diagnostics in a clinical setting. **Expected Impact/Findings:** The study is expected to demonstrate that non-invasive NIR spectroscopy with AI can provide a reliable, risk-free diagnostic alternative for malaria detection in pregnant women. This approach could improve screening and monitoring in sensitive populations, facilitate early treatment, and contribute to malaria control and elimination efforts.

Learned CDQ Questions.

- (1) What are the potential limitations of using a miniaturized 'matchbox size' near-infrared spectrophotometer in field settings for malaria detection, and how does it compare to traditional diagnostic methods such as light microscopy and rapid diagnostic tests?
- (2) Considering the limitations of current rapid diagnostic tests for malaria, what novel strategies could be employed to enhance the accuracy and reliability of NIRS-based diagnostics for other vector-borne diseases?
- (3) How does the sensitivity and specificity of the near-infrared spectroscopy (NIRS) technique for detecting malaria compare to the ultraviolet/visible and near-infrared dual spectroscopic method, particularly for low-level parasitemia?
- (4) How might the morphology-sensitive Raman modes of the malaria pigment hemozoin be leveraged to enhance the detection capabilities of NIRS, and are there known limitations to this approach?
- (5) Given the reported limitations of rapid diagnostic tests (RDTs) and light microscopy, what specific advancements does NIRS offer in terms of accuracy and reliability for malaria diagnosis, especially in low-resource settings?
- (6) How might the dual spectroscopic method combining UV/vis and NIR spectroscopy used for malaria detection be adapted

to detect other blood-borne pathogens with similar parasitemia levels?

- (7) In what ways could the miniaturized 'matchbox size' NIR spectrometer, effective in diagnosing malaria, be applied to other infectious diseases, and what limitations might arise from its current design?
- (8) Given the success of NIRS in detecting malaria parasites, what challenges could be anticipated in adapting this technique to diagnose asymptomatic carriers of other parasitic infections in field settings?
- (9) How does the accuracy of the near-infrared spectroscopy (NIRS) method for detecting malaria parasites compare to the combined ultraviolet-visible and NIR dual spectroscopic method, particularly in detecting low-level ring-staged parasitemia?
- (10) In what ways can the integration of machine learning with NIRS improve the detection accuracy of malaria parasites, and how does this compare to the use of logistic regression in the ultraviolet/visible and near-infrared dual spectroscopic method?

CDQ-Enhanced Ideation (Set B: Detailed Ideas with CDQ).

- **Idea 1: Title:** MIR-Raman Hybrid Diagnostic for Ultra-sensitive Malaria Detection. **Problem:** Current NIR-based methods suffer from inherently low signal-to-noise ratios because NIR overtones are broad and overlapping, making sub-patent detection at the 10^{-7} level extremely difficult. **Objective:** To develop a hybrid diagnostic framework that integrates Mid-Infrared (MIR) fundamental bands with resonant Raman markers to resolve hemozoin signatures. **Hypothesis:** Transitioning from NIR overtones to MIR fundamental vibrational transitions will break the physical sensitivity ceiling and allow for PCR-level sensitivity in field settings. **Method:** Directly addresses the physical sensitivity ceiling by targeting fundamental vibrational transitions. It integrates excitation at $\lambda=633/647\text{nm}$ (Snippet 4) to resolve hemozoin signatures from the bulk hemoglobin background. The framework will utilize portable MIR sensors calibrated with high-resolution Raman data. **Expected Impact/Findings:** The research is expected to produce a field-deployable tool capable of detecting ring-staged malaria with 100% specificity at sub-patent levels, bridging the gap between portable sensors and PCR laboratory accuracy.
- **Idea 2: Title:** Morphology-driven Resonant Raman Assay for Species Differentiation. **Problem:** Identifying specific malaria species at low parasitemia remains a challenge for current non-invasive or portable sensors, which often rely on bulk spectral shifts. **Objective:** To quantify hemozoin dimer structures using resonance Raman to achieve species-level differentiation. **Hypothesis:** Resonant Raman spectroscopy, by mapping crystalline vibrational modes, can distinguish between species-specific hemozoin morphology that is otherwise obscured. **Method:** Apply resonance Raman with excitation at 633 nm and 647 nm to quantify hemozoin dimer structures. This technically grounded

method allows for specific species differentiation by mapping crystalline vibrational modes that are otherwise obscured in broadband spectroscopy. **Expected Impact/Findings:** This assay will enable rapid, species-specific field diagnosis, allowing for targeted treatment protocols in endemic regions where multi-species co-infection is prevalent.

- **Idea 3: Title:** DBS-Based Spectral Calibration Pipeline for Remote Surveillance. **Problem:** The presence of a dominant water absorption peak in the 1000-2500 nm NIR range severely limits the detection of malaria biomarkers in whole blood. **Objective:** To develop a calibration pipeline using Dried Blood Spots (DBS) to remove spectral interference. **Hypothesis:** Utilizing the DBS format will eliminate the water masking effect, significantly increasing the signal-to-noise ratio for parasite-specific absorbance bands. **Method:** Utilizes Dried Blood Spots (DBS) to specifically remove the dominant water absorption peak in the 1000-2500 nm NIR range (Snippet 6), enabling high-fidelity parasite quantification from portable spectrophotometer data. The pipeline includes AI algorithms trained to recognize hemozoin markers in the dehydrated matrix. **Expected Impact/Findings:** This calibration approach will significantly improve the sensitivity of "matchbox size" spectrometers, facilitating global elimination programs through low-cost, high-throughput remote surveillance.

Human Expert Analysis. The expert evaluators significantly preferred the CDQ-enhanced ideas (Set B) due to their rigorous technical grounding. While the baseline ideas provided broad research directions, the CDQ-guided ideas proposed concrete, actionable methodologies that addressed the fundamental physical and biochemical constraints of the domain. For instance, Set B explicitly integrated precise excitation wavelengths ($\lambda=633/647\text{nm}$) and specified the use of Mid-Infrared fundamental bands to overcome the sensitivity barriers of NIR overtones. This transition from conceptual frameworks to executable research plans demonstrates CDQ's unique ability to guide models toward ideas with high technical depth and immediate research viability.

A.3.2 Case Study II: Grounding Protein Disorder Prediction (AIUPred).

Complete Evidence Snippets (Background Context).

- (1) **Snippet 1:** IUPred3: prediction of protein disorder enhanced with unambiguous experimental annotation and visualization of evolutionary conservation. IDPs/IDRs exist without a single well-defined conformation.
- (2) **Snippet 2:** Pfam: The protein families database. A widely used resource for classifying protein sequences into families and domains.
- (3) **Snippet 3:** IUPred2A: context-dependent prediction of protein disorder. Properties of IDRs can depend on various conditions, including binding to partners or redox state.
- (4) **Snippet 4:** DIBS: a repository of disordered binding sites mediating interactions with ordered proteins.
- (5) **Snippet 5:** PhosphoSitePlus: a comprehensive resource for investigating post-translational modifications (PTMs). It encompasses 130,000 non-redundant, manually curated modification sites.

- (6) **Snippet 6:** ELM—the database of eukaryotic linear motifs. Linear motifs play central roles in cell signaling.
- (7) **Snippet 7:** Prediction of Protein Binding Regions in Disordered Proteins.
- (8) **Snippet 8:** Crystal structure of human 53BP1 BRCT domains bound to p53 tumour suppressor.

Baseline Ideation (Set A: Ideas without CDQ).

- **Idea 1: Title:** Integrating Quantum Computing with AlphaFold-Metainference for Enhanced Dynamic Ensemble Prediction of Intrinsically Disordered Proteins. **Problem:** Current computational models struggle to capture the full dynamic landscapes of intrinsically disordered proteins (IDPs), often resulting in static or averaged representations that fail to account for their highly dynamic nature. **Objective:** To develop a quantum-enhanced computational framework that leverages quantum computing capabilities to improve the accuracy and efficiency of dynamic ensemble predictions for IDPs. **Hypothesis:** Quantum computing can enhance the performance of AlphaFold-Metainference by more efficiently exploring the conformational space of IDPs, leading to more accurate and comprehensive dynamic ensemble predictions. **Method:** The study will integrate quantum computing algorithms with the existing AlphaFold-Metainference framework. By utilizing quantum circuits for exploring the IDP conformational space, this approach will aim to achieve a more detailed mapping of dynamic ensembles. Performance will be benchmarked against classical computational methods using a dataset of experimentally characterized IDPs. **Expected Impact/Findings:** The integration of quantum computing is expected to significantly enhance the accuracy and computational efficiency of IDP dynamic ensemble predictions. This advancement could lead to better understanding of IDP functions and interactions, providing new insights for therapeutic targeting and synthetic biology applications.
- **Idea 2: Title:** Integrative Modeling of Intrinsically Disordered Protein Landscapes Using Hybrid AI-Physics Pipelines. **Problem:** Current computational models struggle to capture the full dynamic landscapes of intrinsically disordered proteins (IDPs), which limits our understanding of their functional roles in biological systems. **Objective:** To develop an integrative computational framework that combines AI-driven generative models with physics-based simulations to accurately predict the dynamic conformational ensembles of IDPs and their functional interactions. **Hypothesis:** A hybrid modeling approach that leverages machine learning, specifically transformer-based diffusion models, in conjunction with molecular dynamics simulations can provide a more comprehensive and accurate representation of the conformational and functional landscapes of IDPs. **Method:** 1. Develop a hybrid AI-physics pipeline that integrates a transformer-based diffusion model (e.g., IDPForge) with molecular dynamics-informed ensemble simulations (e.g., AlphaFold-Metainference). 2. Use this pipeline to generate detailed dynamic ensembles for a set of benchmark

IDPs with known experimental data. 3. Validate the predictions against experimental data from resources like PhosphoSitePlus and DIBS. 4. Assess the functional relevance of predicted ensembles by examining their ability to predict known interactions and post-translational modifications. 5. Conduct case studies on disease-related IDPs to test the pipeline's applicability in therapeutic contexts. **Expected Impact/Findings:** This research is expected to significantly enhance our ability to model dynamic IDP ensembles with high fidelity, providing insights into their functional mechanisms and potential therapeutic targets. The hybrid pipeline could set new standards for benchmarking disorder predictions and facilitate the design of IDP-targeted drugs and synthetic biology applications.

- **Idea 3: Title:** Integrative Framework for Functional Annotation of Intrinsically Disordered Protein Ensembles. **Problem:** Despite advances in modeling the structural ensembles of intrinsically disordered proteins (IDPs), there remains a significant gap in linking these dynamic conformations to their biological functions, particularly in signaling and regulatory processes. **Objective:** To develop a comprehensive framework that integrates dynamic ensemble data with functional annotations to predict and validate the interaction networks and roles of IDPs in cellular processes. **Hypothesis:** The dynamic conformational landscapes of IDPs can be effectively linked to specific functional outcomes by leveraging a combination of machine learning and graph-based interaction modeling, enhanced by experimental validation. **Method:** The research will employ a hybrid approach combining machine learning models trained on existing IDP ensemble data with graph-based interaction algorithms to predict potential functional roles. This will be complemented by experimental validation using a high-throughput binding assay to assess predicted interactions and functions in vitro and in cell-based systems. The framework will also incorporate evolutionary conservation data and post-translational modification insights to refine predictions. **Expected Impact/Findings:** The study is anticipated to provide a robust methodology for assigning functional roles to IDPs based on their dynamic conformational ensembles, bridging the gap between structural predictions and biological function. This could lead to novel insights into IDP-mediated signaling pathways and offer new therapeutic targets by elucidating the functional mechanisms of disordered proteins in disease contexts.

Learned CDQ Questions.

- (1) In what ways does the AIUPred web server provide enhanced visual analysis capabilities compared to previous versions, and how do these capabilities facilitate the understanding of evolutionary conservation in intrinsically disordered proteins?
- (2) What are the potential implications of AIUPred's ability to analyze whole genomes rapidly on the study of protein-protein interactions mediated by intrinsically disordered

proteins, especially in the context of the DIBS repository's lack of large-scale interaction data?

- (3) How does the AIUPred model incorporate data from existing resources like PhosphoSitePlus and ELM to improve the prediction of post-translational modifications and linear motifs in disordered proteins, and what gaps remain in understanding their functional roles?
- (4) Given the comprehensive data on post-translational modifications in PhosphoSitePlus, how can AIUPred incorporate this information to refine predictions of protein disorder, and what novel methodologies would be required for this integration?
- (5) What are the specific challenges in characterizing intrinsically disordered protein regions (IDRs) using computational methods, and how does AIUPred overcome these compared to other methods like IUPred3?
- (6) How does AIUPred's approach to energy estimation differ from traditional methods, and what evidence is there to suggest this leads to more accurate predictions of protein disorder?
- (7) How does AIUPred address the context-dependent nature of protein disorder, such as changes in redox state or protein binding, and what evidence supports its effectiveness in these scenarios?
- (8) Given the comprehensive data on post-translational modifications in PhosphoSitePlus, how can AIUPred incorporate this information to refine predictions of protein disorder, and what novel methodologies would be required for this integration?
- (9) How does AIUPred's integration of deep learning techniques compare to other machine learning approaches in terms of predicting the disorder of proteins, and what specific evidence supports its superiority?
- (10) What are the limitations of AIUPred in predicting protein disorder in comparison to other tools like IUPred3, and how could these limitations be addressed?

CDQ-Enhanced Ideation (Set B: Detailed Ideas with CDQ).

- **Idea 1: Title:** Integrating Evolutionary Conservation and Post-Translational Modifications for Enhanced Disorder Prediction in Proteins. **Problem:** Current computational models for predicting intrinsically disordered regions (IDRs) in proteins lack the incorporation of evolutionary conservation metrics and post-translational modifications (PTMs) as features, limiting the understanding of functionally conserved disordered regions and their regulatory mechanisms. **Objective:** To develop an advanced computational model that integrates evolutionary conservation data and PTM information to improve the accuracy and contextual relevance of IDR predictions. **Hypothesis:** Incorporating evolutionary conservation metrics along with PTM data into a disorder prediction model will enhance its ability to identify functionally significant disordered regions and provide insights into their regulatory roles. **Method:** 1. Collect and integrate datasets containing evolutionary conservation scores, PTM sites (from PhosphoSitePlus), and linear motifs (from ELM) with existing protein disorder annotations. 2.

Develop a transformer-based neural network model that includes these features as input, alongside traditional sequence data. 3. Train and validate the model using benchmark datasets and compare its performance against existing tools like AIUPred and IUPred3. 4. Analyze the model's predictions to identify key conserved disordered regions and their associated PTMs, testing their functional implications in biological processes. **Expected Impact/Findings:** The integration of evolutionary conservation and PTM data into disorder prediction models is expected to improve prediction accuracy and provide a deeper understanding of the functional significance of IDRs. This could lead to new insights into protein regulation, interactions, and potential therapeutic targets, particularly in complex biological systems where disordered regions play critical roles.

- **Idea 2: Title:** Integrating Multi-Modal Data for Enhanced Prediction of Functional Intrinsically Disordered Regions. **Problem:** Current tools for predicting intrinsically disordered regions (IDRs) in proteins lack integrated multi-modal data inputs, such as post-translational modifications (PTMs) and linear motifs, limiting the contextual understanding of their functional roles. **Objective:** To develop a prediction model that incorporates PTM, linear motif, and evolutionary conservation data to improve the accuracy and functional contextualization of IDR predictions. **Hypothesis:** Incorporating multi-modal data inputs, including PTMs, motifs, and conservation metrics, into a predictive model will enhance the accuracy and functional relevance of IDR predictions, providing better insights into protein function and regulation. **Method:** Develop a neural network-based model that integrates PTM and motif data from resources like PhosphoSitePlus and ELM, alongside evolutionary conservation metrics. Use transformer-based energy embeddings and train the model on a diverse dataset of experimentally verified IDRs to evaluate improvements in prediction accuracy and functional annotation. **Expected Impact/Findings:** The enhanced model is expected to provide more accurate and contextually relevant predictions of IDRs, facilitating the discovery of novel IDP-mediated interactions and regulatory mechanisms. This approach could lead to better understanding and targeting of IDRs in therapeutic contexts, advancing both basic research and applied biotechnology.
- **Idea 3: Title:** Integrating Post-Translational Modification and Linear Motif Data into AIUPred for Enhanced Contextual Prediction of Intrinsically Disordered Regions. **Problem:** Current AIUPred models lack integration of post-translational modifications (PTMs) and linear motif data as features, limiting the contextual accuracy of disorder and binding predictions. **Objective:** To develop an enhanced AIUPred model that incorporates PTM and linear motif data as integral features to improve the prediction of functional disordered regions under various biological contexts. **Hypothesis:** Incorporating PTM and linear motif data into AIUPred's predictive model will increase the accuracy of identifying functionally relevant disordered regions, particularly in the context of protein regulation and signaling.

Method: Develop a machine learning framework incorporating PTM and linear motif data from PhosphoSitePlus and ELM databases as features in AIUPred's transformer-based model. Conduct comparative analysis using CAID benchmarks to evaluate performance improvements in predicting disordered regions and their functional roles. **Expected Impact/Findings:** The enhanced model is anticipated to provide more accurate predictions of IDRs, particularly under different biological contexts such as signaling and regulation. This could lead to better understanding of IDP-mediated interactions and facilitate the discovery of novel therapeutic targets.

Human Expert Analysis. The expert evaluation confirmed that the CDQ-enhanced ideas (Set B) achieved much higher feasibility scores by replacing speculative or theoretically distant concepts with structured, data-grounded methodologies. In Case Study II, whereas the baseline ideas proposed high-risk directions such as "Quantum Computing" or general "AI-Physics hybrid pipelines" without specific execution paths, the CDQ-guided ideas directly addressed the "Evidence Gap." By answering the questions regarding the 130,000 modification sites in **PhosphoSitePlus (Snippet 5)**², the engines proposed implementable architectures like PTM-tokenized transformers. This shift from hypothetical "black-box" approaches to evidence-driven implementations is the primary driver behind the superior expert preference for Set B.

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